Transolver: A Fast Transformer Solver for PDEs on General Geometries

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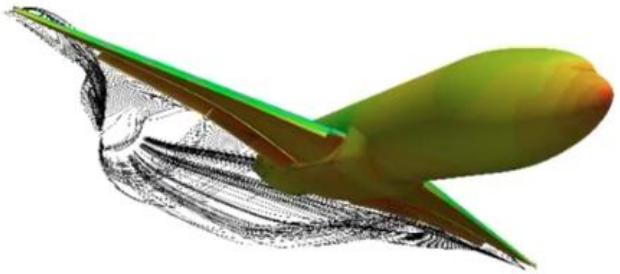
参考：Transolver to Transolver3
https://www.youtube.com/watch?v=oef6G3kSQcg&t=3430s

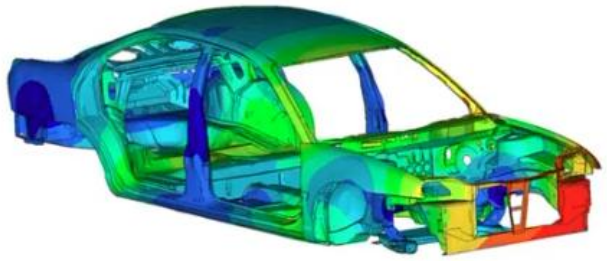
2026.10.15张欣宇

PDESolver 的求解背景：

（1）数值求解

物理场可由PDE方差描述







$$\frac{\partial v_x}{\partial x}+\frac{\partial v_y}{\partial y}+\frac{\partial v_z}{\partial z}=0$$

$$\rho\left(\frac{\partial v_x}{\partial t}+v_x\frac{\partial v_x}{\partial x}+v_y\frac{\partial v_x}{\partial y}+v_z\frac{\partial v_x}{\partial z}\right)=-\frac{\partial P}{\partial x}+\mu\left(\frac{\partial^2 v_x}{\partial x^2}+\frac{\partial^2 v_x}{\partial y^2}+\frac{\partial^2 v_x}{\partial z^2}\right)+\rho g_x$$

$$\rho\left(\frac{\partial v_y}{\partial t}+v_x\frac{\partial v_y}{\partial x}+v_y\frac{\partial v_y}{\partial y}+v_z\frac{\partial v_y}{\partial z}\right)=-\frac{\partial P}{\partial y}+\mu\left(\frac{\partial^2 v_y}{\partial x^2}+\frac{\partial^2 v_y}{\partial y^2}+\frac{\partial^2 v_y}{\partial z^2}\right)+\rho g_y$$

$$\rho\left(\frac{\partial v_z}{\partial t}+v_x\frac{\partial v_z}{\partial x}+v_y\frac{\partial v_z}{\partial y}+v_z\frac{\partial v_z}{\partial z}\right)=-\frac{\partial P}{\partial z}+\mu\left(\frac{\partial^2 v_z}{\partial x^2}+\frac{\partial^2 v_z}{\partial y^2}+\frac{\partial^2 v_z}{\partial z^2}\right)+\rho g_z$$

3-D Navier-Stokes equations

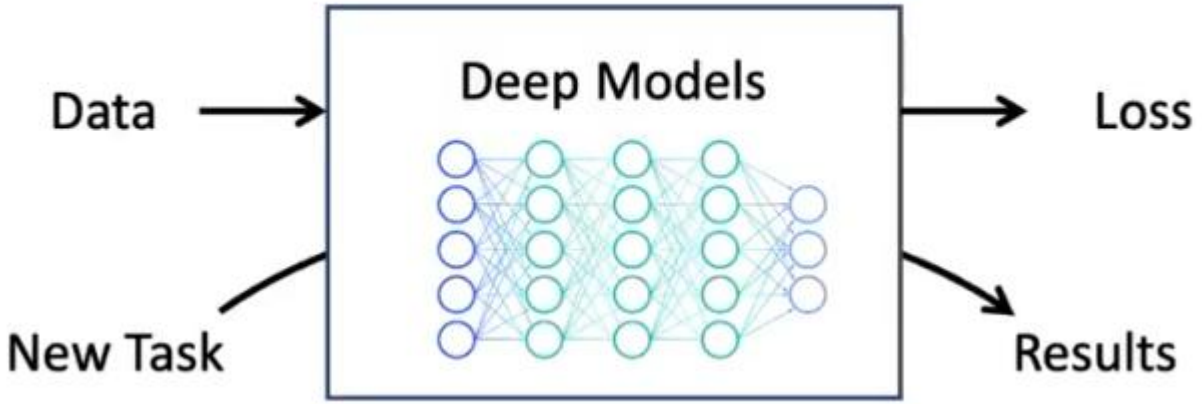
$$\varepsilon_{xx}=\frac{\partial u_x}{\partial x},\quad \varepsilon_{yy}=\frac{\partial u_y}{\partial y},\quad \varepsilon_{zz}=\frac{\partial u_z}{\partial z}$$

$$\varepsilon_{xy}=\frac{1}{2}\left(\frac{\partial u_x}{\partial y}+\frac{\partial u_y}{\partial x}\right),\quad \varepsilon_{xz}=\frac{1}{2}\left(\frac{\partial u_x}{\partial z}+\frac{\partial u_z}{\partial x}\right),\quad \varepsilon_{yz}=\frac{1}{2}\left(\frac{\partial u_y}{\partial z}+\frac{\partial u_z}{\partial y}\right)$$

3-D Stress-Strain relations

PDESolver 的求解背景：

（2）Neural Network



统一模型

$$\mathcal{L}^{\text{physics}} = \mathbb{E}_{\mathcal{D}}\left[\|\mathcal{F}_\theta(\mathbf{x}; G, S) - \boldsymbol{u}(\mathbf{x})\|_2^2\right].$$

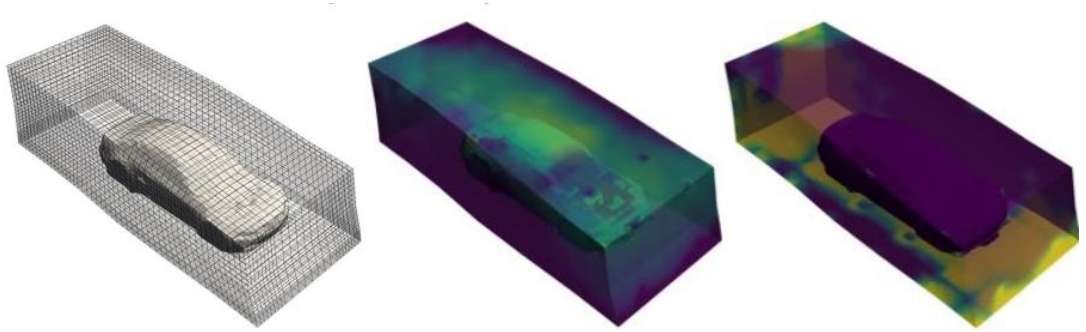
模型预测：给定几何 ***G***、物理场条件（流速、方向等）***S***，在点 ***x***处的物理量

真实物理量（数值求解器算出来的）

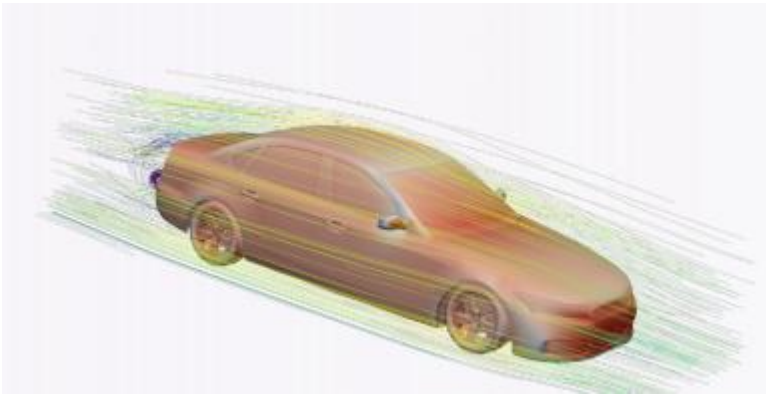
一步到位学习**最终结果**

PDESolver 的求解任务形式

$$\mathcal{L}^{\text{physics}} = \mathbb{E}_{\mathcal{D}} \left[\| \mathcal{F}_\theta(\mathbf{x}; G, S) - \boldsymbol{u}(\mathbf{x}) \|_2^2 \right].$$

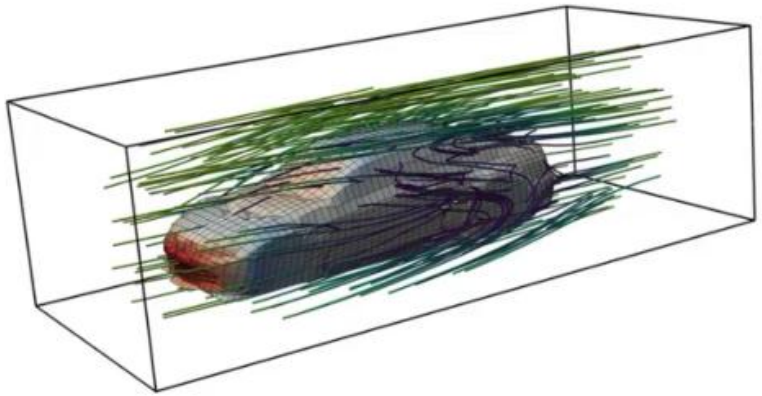


- **G**：一辆车的表面网格，来自 ShapeNet。
- **S**：72 km/h 的迎风，所有样本相同，所以**根本没有作为输入出现**。
- **x**：这辆车周围的 32186 个点，包括车表面点和周围空气里的点。
- **u(x)**：每个点上的速度向量和压力。



输入：入一辆车：[1, 32186, 7]每个点 7 维 [x, y, z, sdf, nx, ny, nz]
输出：[1, 32186, 4]，每个点是 (u, v, w, p)

Challenges in Practical Industrial Design



Task: Estimate the drag coefficient of a given shape:

Surrounding Wind & Surface Pressure

1. Large-scale meshes → **Huge computation cost** → 工业级 CFD 仿真需要**百万级甚至千万级网格点**
2. Complex and unstructured geometrics → **Complex geometric learning**
3. Naiver-Stokes equation → **Intricate physical correlations**

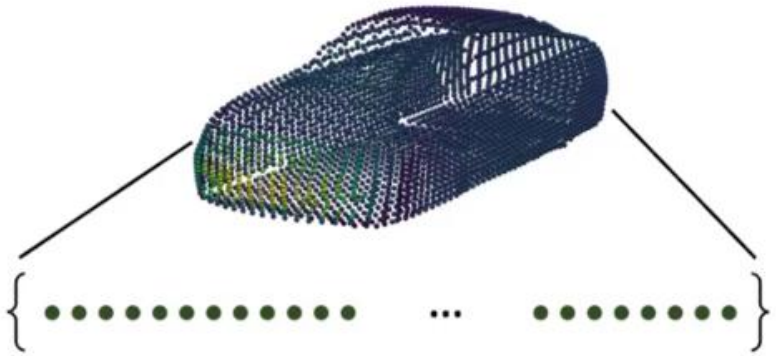
CNN：需要规则网格；

GNN：复杂几何上构建邻接图很麻烦。

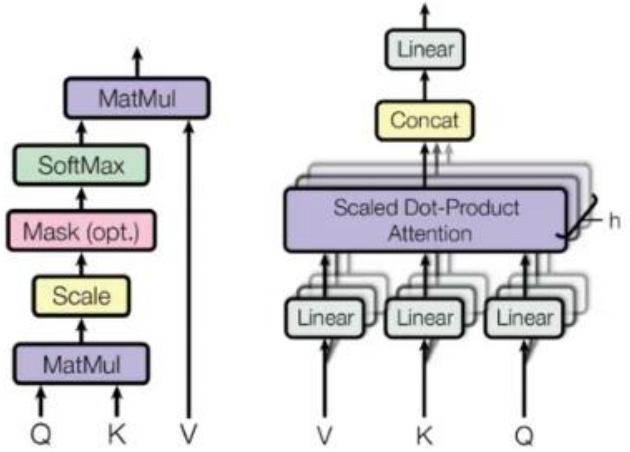
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物理场全局耦合——车头的压力波会影响车尾的涡流

基于Transformer的PDESolver



不需要规则网格，不需要邻接图。



(1) Geometries as point sequences (2) Attention as Monte Carlo Integral

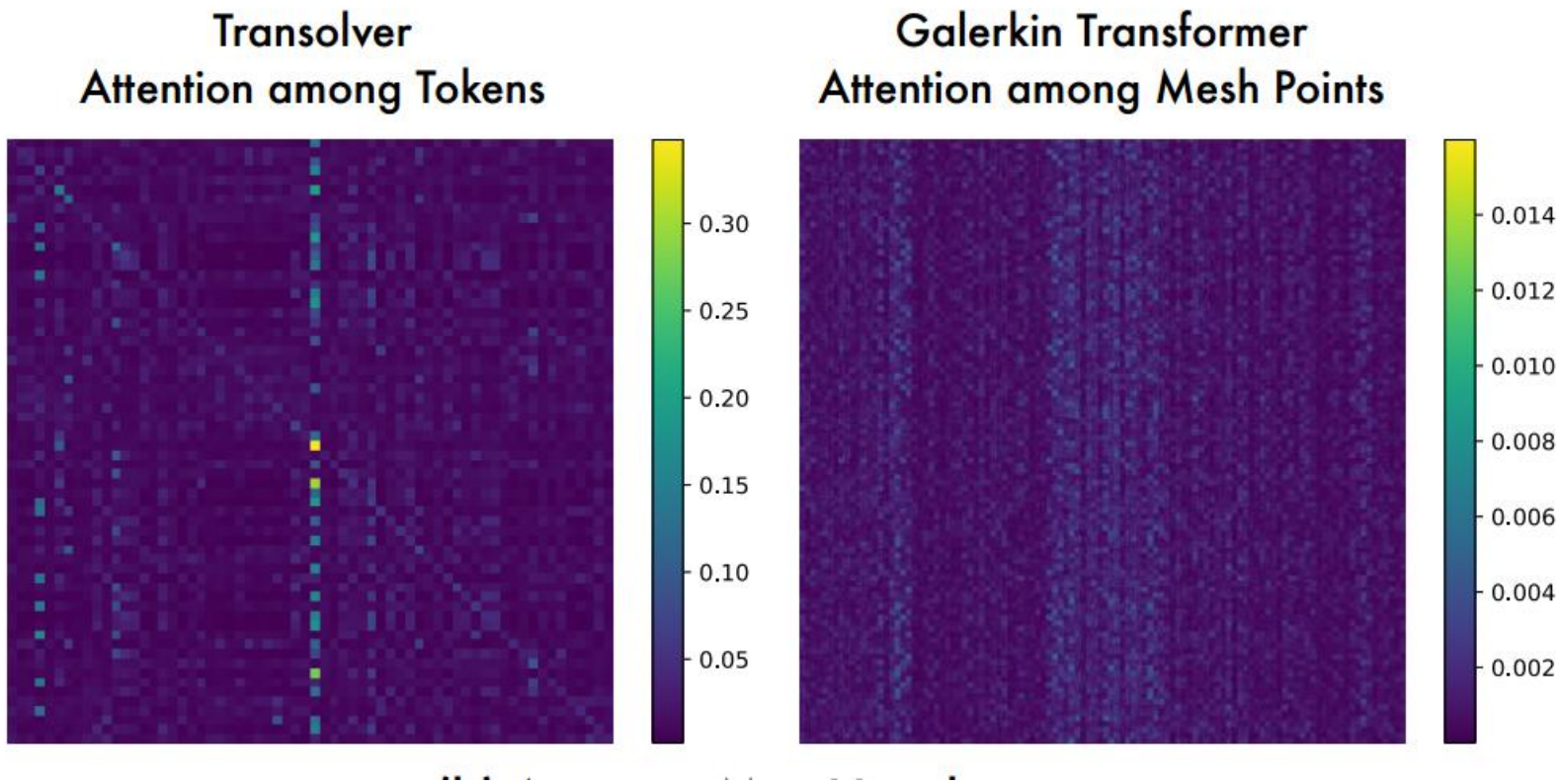
$$\text{Attention}(Q,K,V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d}}\right)V$$

$$\mathcal{G}(u)(g^*) \approx \sum_{j=1}^{N} \kappa(g^*, g_j)u(g_j)$$

非局部耦合：空间上远离的点，通过积分核相互影响。
用 N个采样点的加权和，去近似一个连续的空间积分。

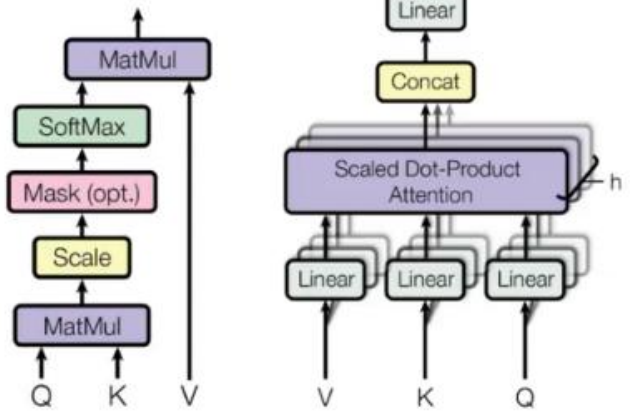
好处：能直接处理点云、不规则网格

- 问题：复杂度 $O(N^2)$
- 网格点多的时候难以捕捉相关性


Transolver
Attention among Tokens
0.30
0.25
0.20
0.15
0.10
0.05
Galerkin Transformer
Attention among Mesh Points
0.014
0.012
0.010
0.008
0.006
0.004
0.002

Transolver: A Fast Transformer Solver for PDEs on General Geometries





主要思想：在离散网格坐标上去建模高效的物理状态关联
（基于Transformer架构实现PDE求解）



Discretized Domain

Previous Work

Being “trapped” to superficial and unwieldy meshes

经典视觉Patch?

物理场海量自由度由少量模态表达

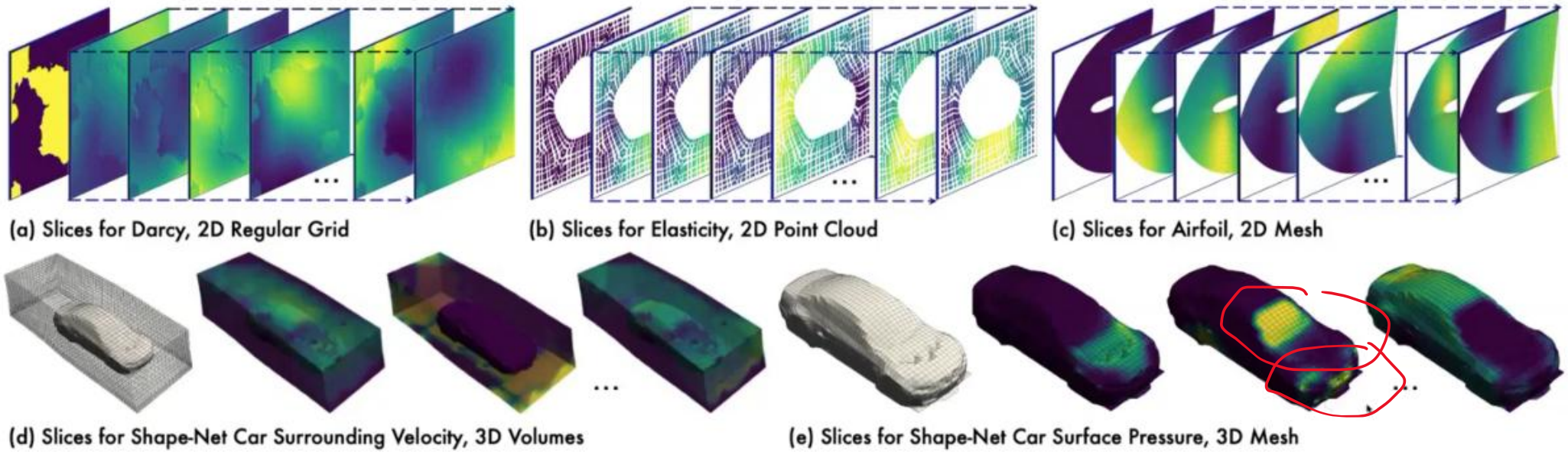
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真正想做的：如何对大量的Mesh做有效的Tokenizer

↓

Dynamic Slicing

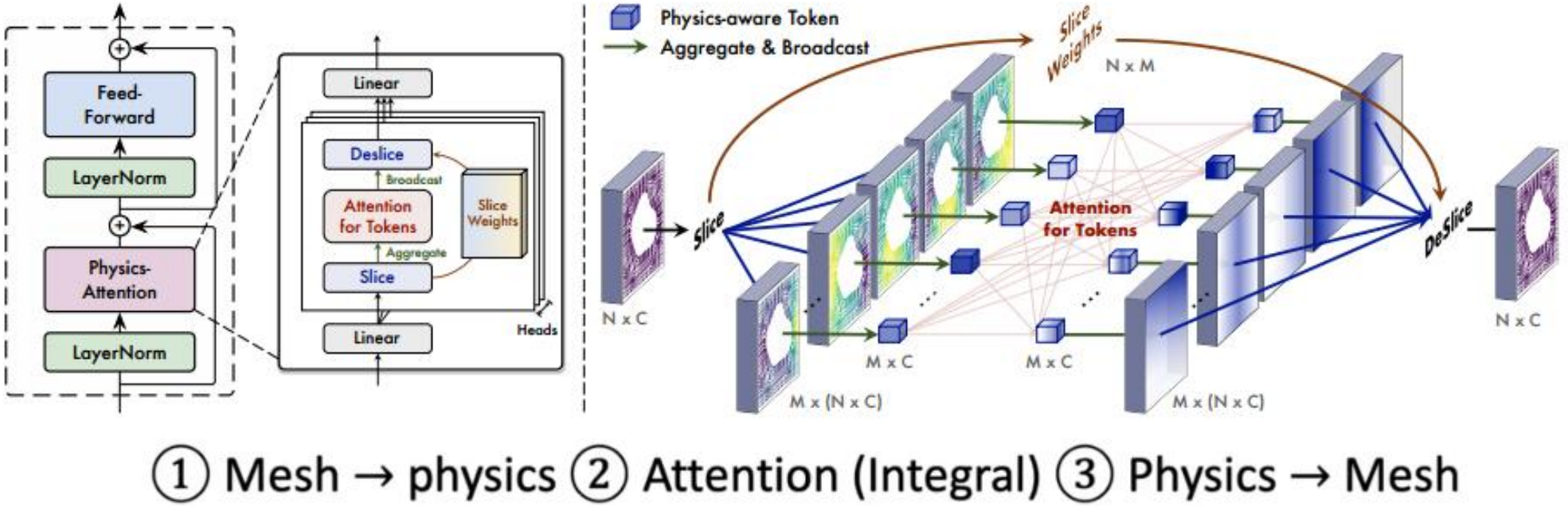
Intuition——Dynamic Slice



物理直觉：Physical states(而不是空间相近)相近的点应该处于同一个Slice切片然后encode到token中

Physical states：不对应真实物理量，而是学习物理特征得到的某些相关模式

Physics-Attention



① Mesh → physics ② Attention (Integral) ③ Physics → Mesh

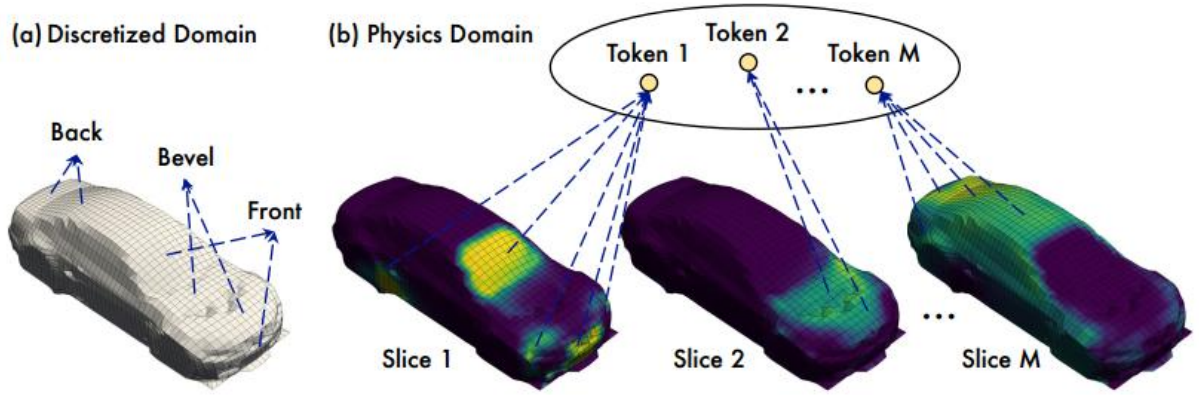
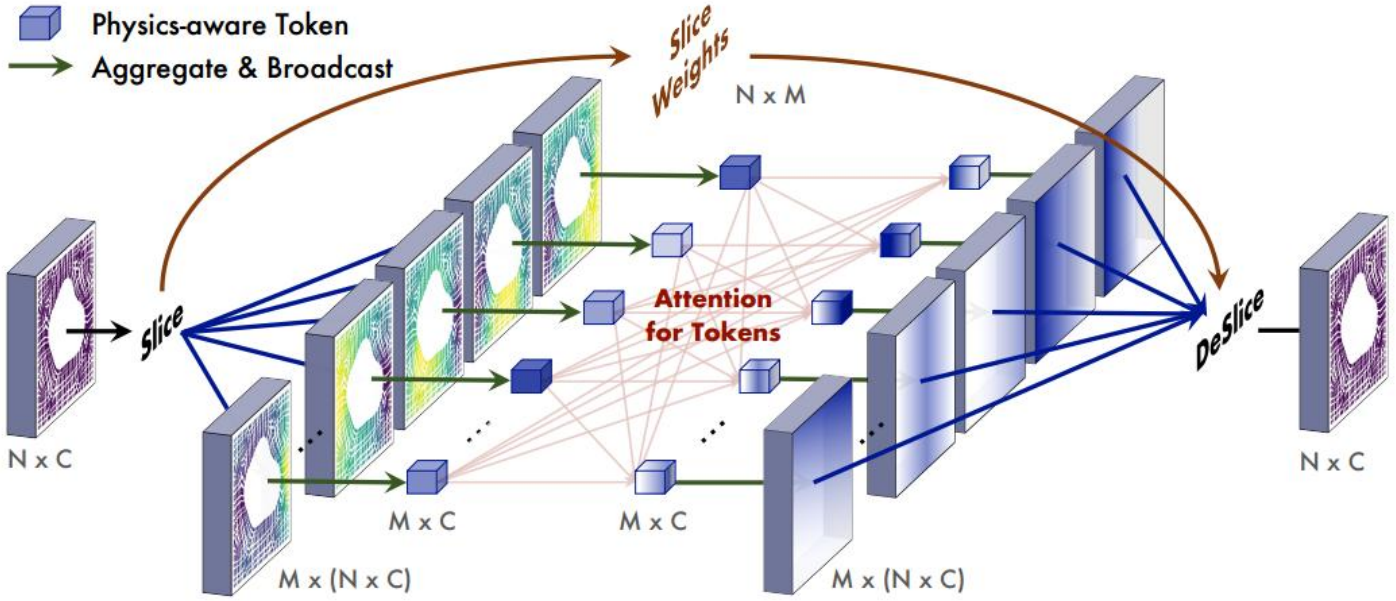
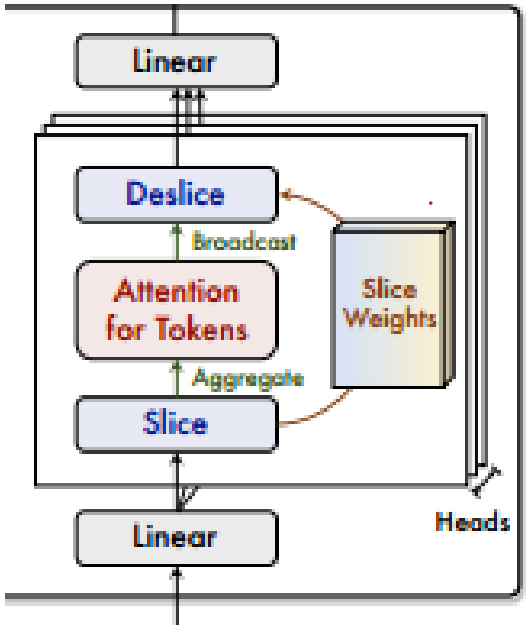


Figure 2. Learning physics-aware tokens from Transolver slices.

Method

把 N个点压缩成 M个物理 Token







$$\{\mathbf{w}_i\}_{i=1}^N = \{\mathrm{Softmax}\left(\mathrm{Project}\left(\mathbf{x}_i\right)\right)\}_{i=1}^N$$

$$\mathbf{s}_j = \{\mathbf{w}_{i,j}\mathbf{x}_i\}_{i=1}^N,$$

$$\mathbf{z}_j = \frac{\sum_{i=1}^N \mathbf{s}_{j,i}}{\sum_{i=1}^N \mathbf{w}_{i,j}} = \frac{\sum_{i=1}^N \mathbf{w}_{i,j}\mathbf{x}_i}{\sum_{i=1}^N \mathbf{w}_{i,j}}$$

加权和

权重和

输入：$\mathbf{x} \in \mathbb{R}^{N\times C_{\text{in}}}$

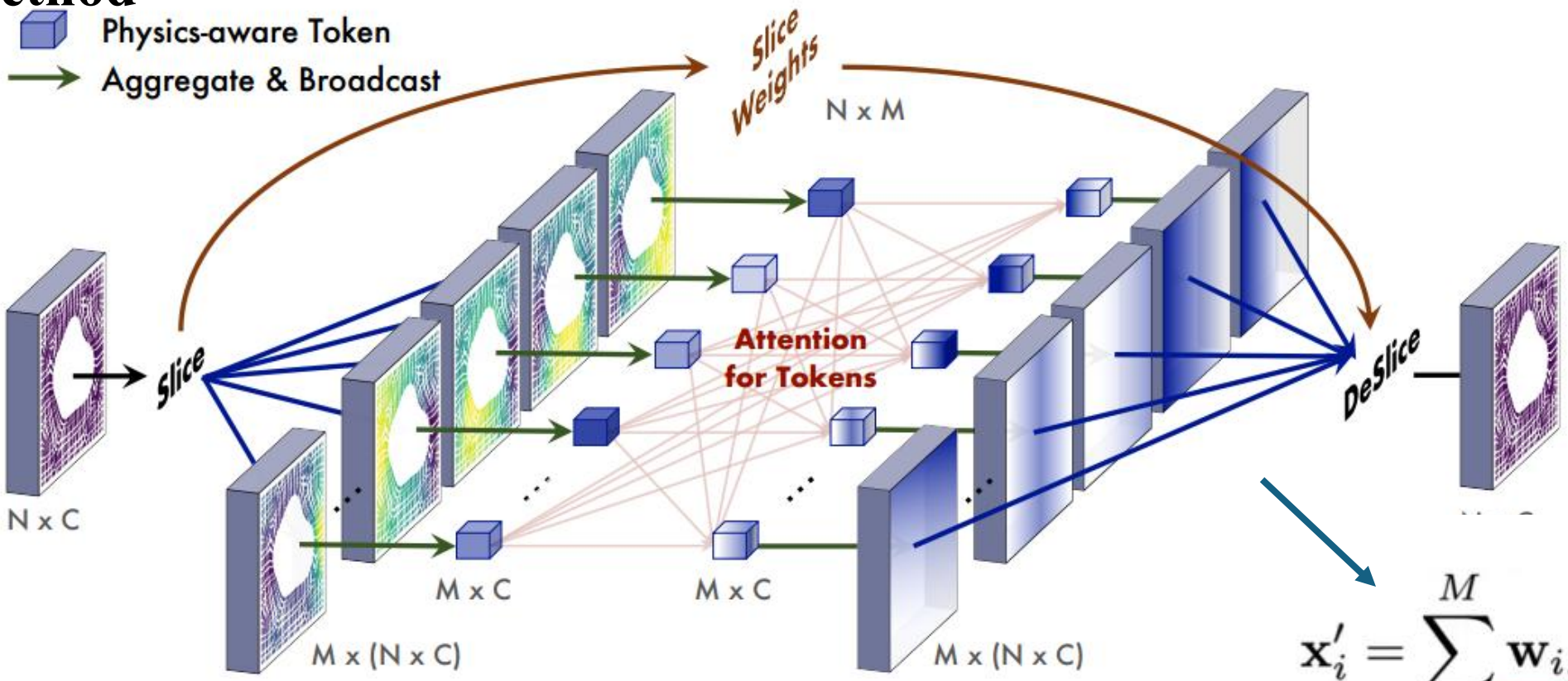
Project 输出：$\mathbb{R}^{N\times M}$

Softmax 输出：$\mathbf{w} \in \mathbb{R}^{N\times M}$

$\mathbf{s} \in \mathbb{R}^{M\times C_{\text{in}}}$第 j个切片的加权和（未归一化）

全局归一化：每个切片的 Token 是点的加权平均，不受切片大小影响

Method



$$\mathbf{x}_i' = \sum_{j=1}^{M} \mathbf{w}_{i,j} \mathbf{z}_j'$$

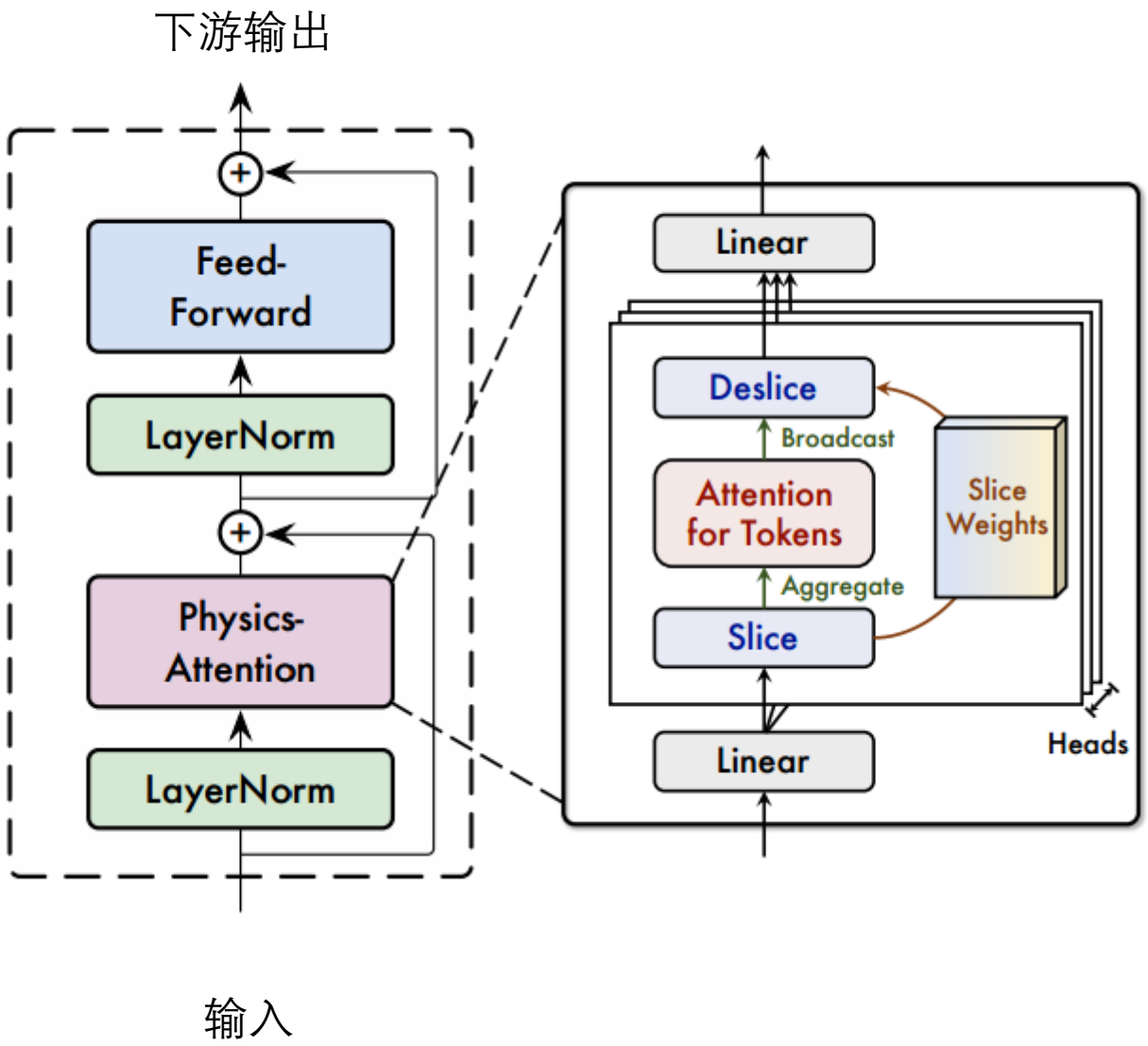
Slice weight

$$\mathbf{q}, \mathbf{k}, \mathbf{v} = \mathrm{Linear}(\mathbf{z}), \quad \mathbf{z}' = \mathrm{Softmax}\left(\frac{\mathbf{q}\mathbf{k}^{\mathsf{T}}}{\sqrt{C}}\right)\mathbf{v}$$

1. Complexity: $\mathcal{O}(N^2C) \rightarrow \mathcal{O}(M^2C)$
2. Capture interactions among physics states

Method:

最终网络构成



Experiment Results

22% error reduce

Model	Point Cloud	Structured Mesh			Regular Grid	
	Elasticity	Plasticity	Airfoil	Pipe	Navier–Stokes	Darcy
FNO (Li et al., 2021)	/	/	/	/	0.1556	0.0108
WMT (Gupta et al., 2021)	0.0359	0.0076	0.0075	0.0077	0.1541	0.0082
U-FNO (Wen et al., 2022)	0.0239	0.0039	0.0269	0.0056	0.2231	0.0183
Geo-FNO (Li et al., 2022)	0.0229	0.0074	0.0138	0.0067	0.1556	0.0108
U-NO (Rahman et al., 2023)	0.0258	0.0034	0.0078	0.0100	0.1713	0.0113
F-FNO (Tran et al., 2023)	0.0263	0.0047	0.0078	0.0070	0.2322	0.0077
LSM (Wu et al., 2023)	0.0218	0.0025	0.0059	0.0050	0.1535	0.0065
Galerkin (Cao, 2021)	0.0240	0.0120	0.0118	0.0098	0.1401	0.0084
HT-Net (Liu et al., 2022)	/	0.0333	0.0065	0.0059	0.1847	0.0079
OFormer (Li et al., 2023c)	0.0183	0.0017	0.0183	0.0168	0.1705	0.0124
GNOT (Hao et al., 2023)	0.0086	0.0336	0.0076	0.0047	0.1380	0.0105
FactFormer (Li et al., 2023d)	/	0.0312	0.0071	0.0060	0.1214	0.0109
ONO (Xiao et al., 2024)	0.0118	0.0048	0.0061	0.0052	0.1195	0.0076
Transolver (Ours)	**0.0064**	**0.0012**	**0.0053**	**0.0033**	**0.0900**	**0.0057**
Relative Promotion	25.6%	29.4%	10.2%	29.7%	24.7%	12.3%

Benchmarks

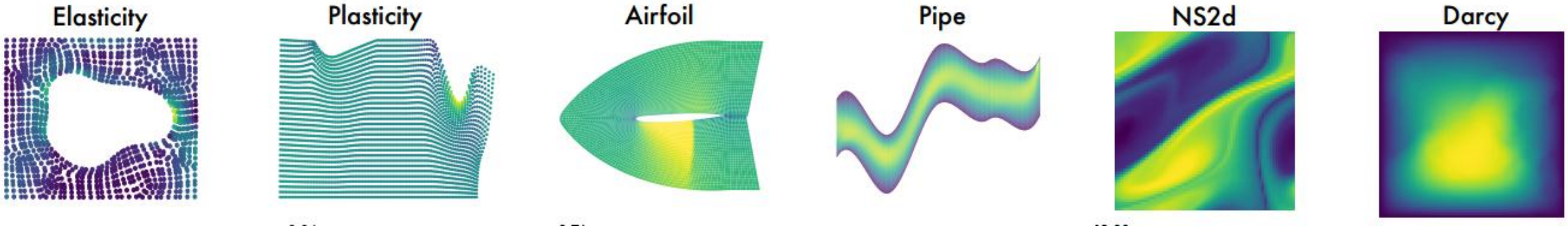
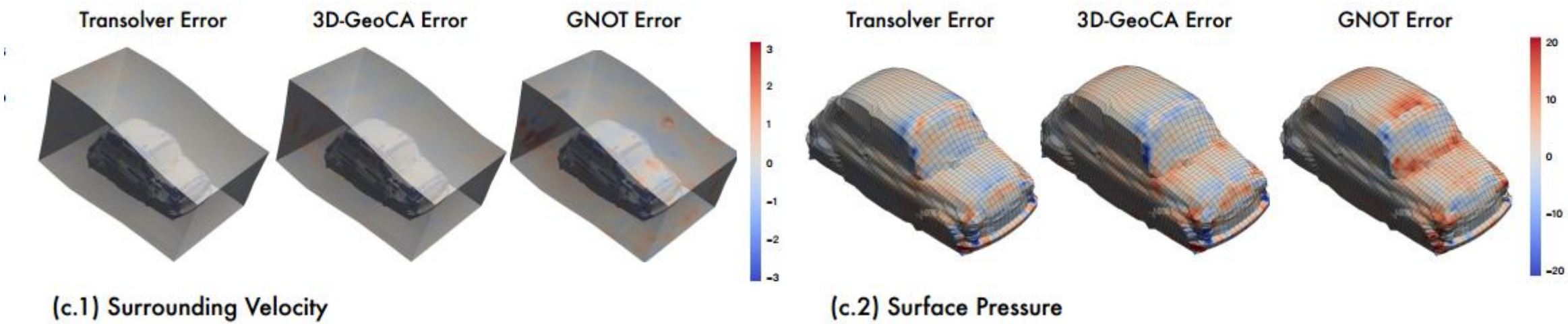


Table 3. Performance comparison on practical design tasks. Both benchmarks are with unstructured mesh. In addition to the relative L2 of the surrounding (Volume) and surface (Surf) physics fields, the relative L2 of drag coefficient (C_D) and lift coefficient (C_L) is also recorded, along with their Spearman's rank correlations ρ_D and ρ_L. A Spearman's correlation close to 1 indicates better performance.

Model*	Shape-Net Car				AirfRANS			
	Volume ↓	Surf ↓	C_D ↓	ρ_D ↑	Volume ↓	Surf ↓	C_L ↓	ρ_L ↑
Simple MLP	0.0512	0.1304	0.0307	0.9496	0.0081	0.0200	0.2108	0.9932
GraphSAGE (Hamilton et al., 2017)	0.0461	0.1050	0.0270	0.9695	0.0087	0.0184	<u>0.1476</u>	<u>0.9964</u>
PointNet (Qi et al., 2017)	0.0494	0.1104	0.0298	0.9583	0.0253	0.0996	0.1973	0.9919
Graph U-Net (Gao & Ji, 2019)	0.0471	0.1102	0.0226	0.9725	0.0076	0.0144	0.1677	0.9949
MeshGraphNet (Pfaff et al., 202		781	0.0168	0.9840	0.0214	0.0387	0.2252	0.9945
GNO (Li et al., 2020a)	0.0383	0.0815	0.0172	0.9834	0.0269	0.0405	0.2016	0.9938
Galerkin (Cao, 2021)	0.0339	0.0878	0.0179	0.9764	0.0074	0.0159	0.2336	0.9951
geo-FNO (Li et al., 2022)	0.1670	0.2378	0.0664	0.8280	0.0361	0.0301	0.6161	0.9257
GNOT (Hao et al., 2023)	0.0329	0.0798	0.0178	0.9833	<u>0.0049</u>	<u>0.0152</u>	0.1992	0.9942
GINO (Li et al., 2023a)	0.0386	0.0810	0.0184	0.9826	0.0297	0.0482	0.1821	0.9958
3D-GeoCA (Deng et al., 2024)	<u>0.0319</u>	<u>0.0779</u>	<u>0.0159</u>	<u>0.9842</u>	/	/	/	/
Transolver (Ours)	**0.0207**	**0.0745**	**0.0103**	**0.9935**	**0.0037**	**0.0142**	**0.1030**	**0.9978**

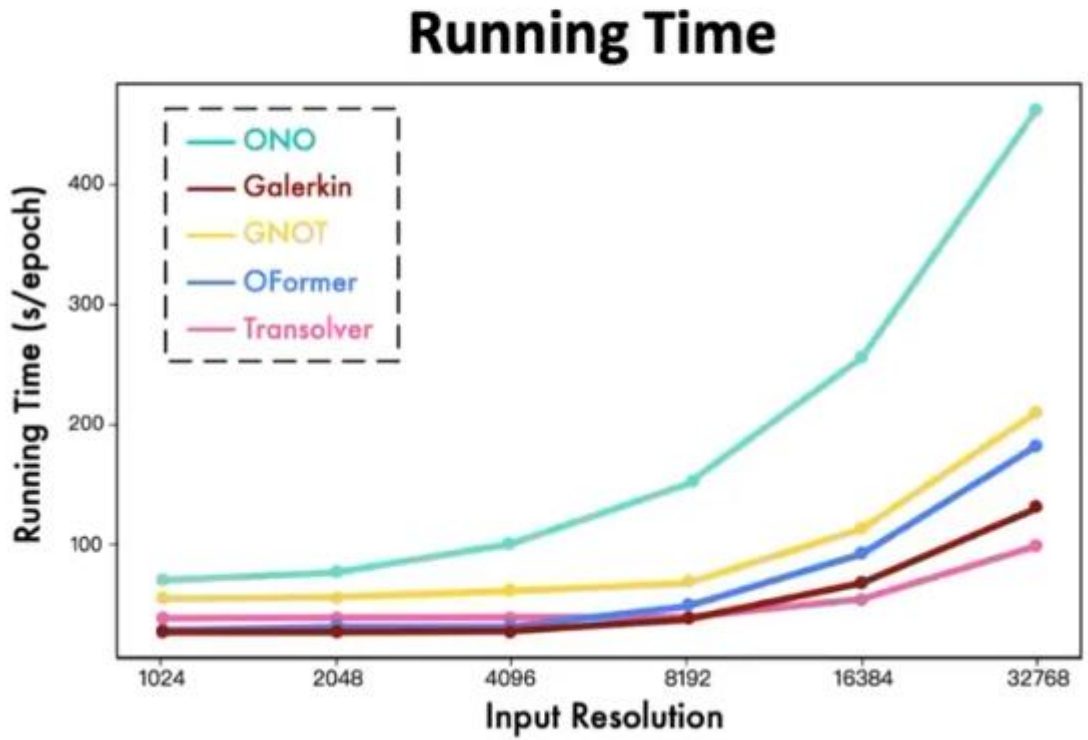
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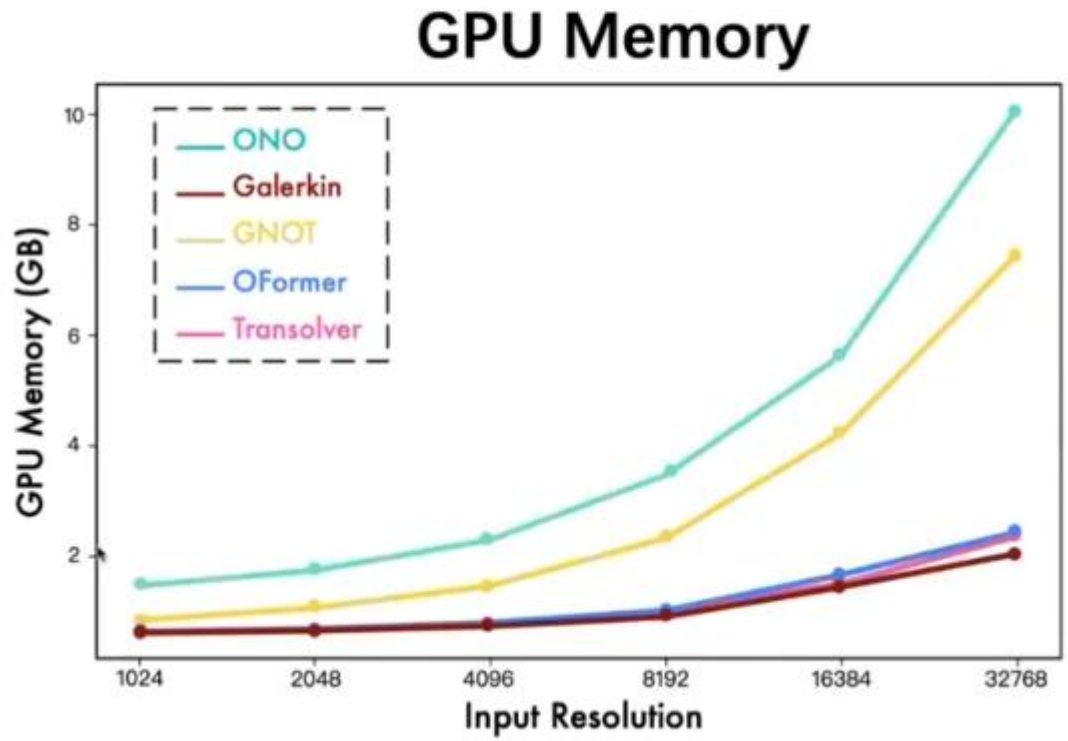


(c.1) Surrounding Velocity

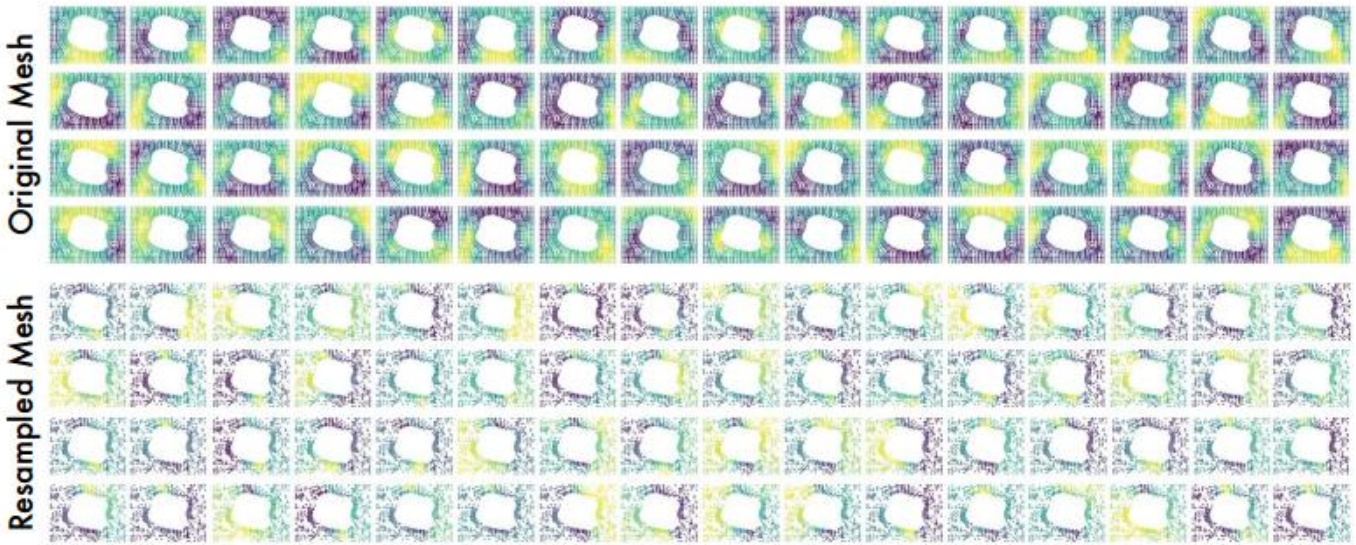
(c.2) Surface Pressure

Efficiency

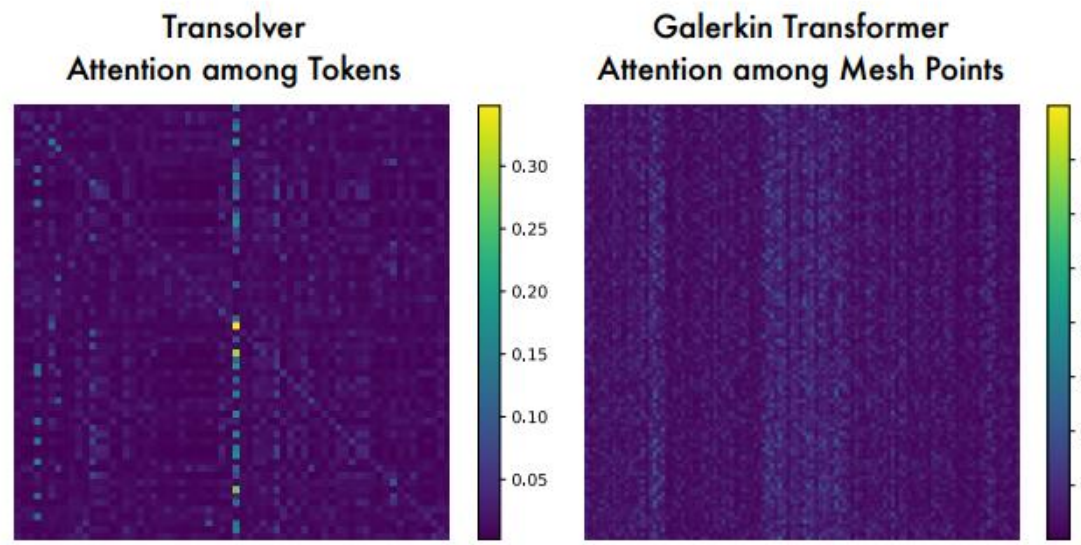




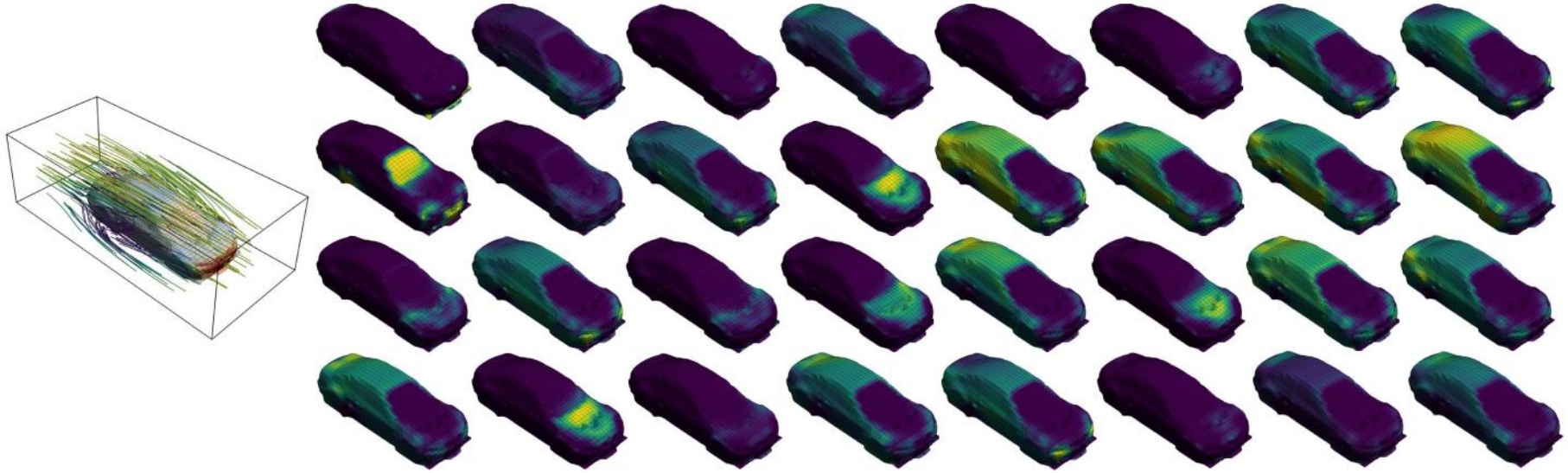
Transolver is faster than linear Transformers in large-scale meshes.



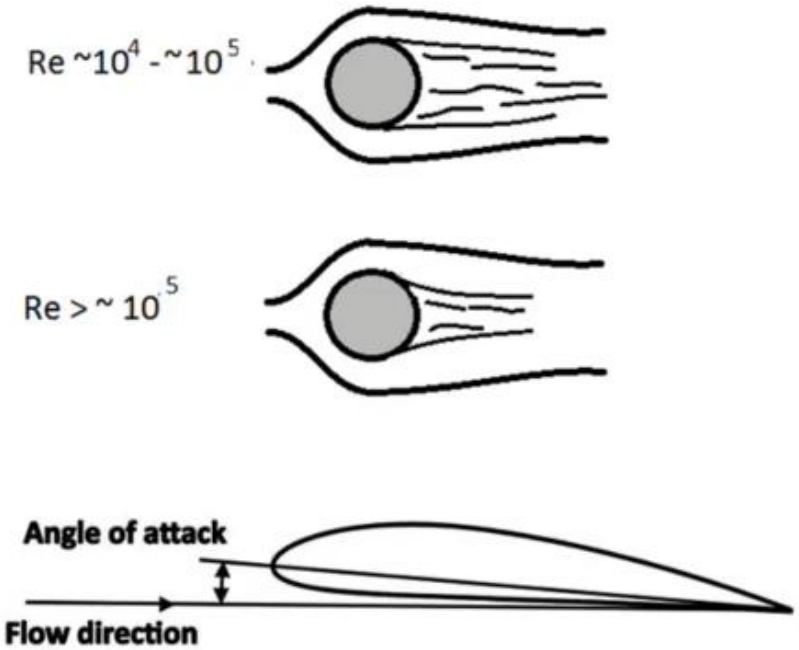
(a) Learned Slice Visualization



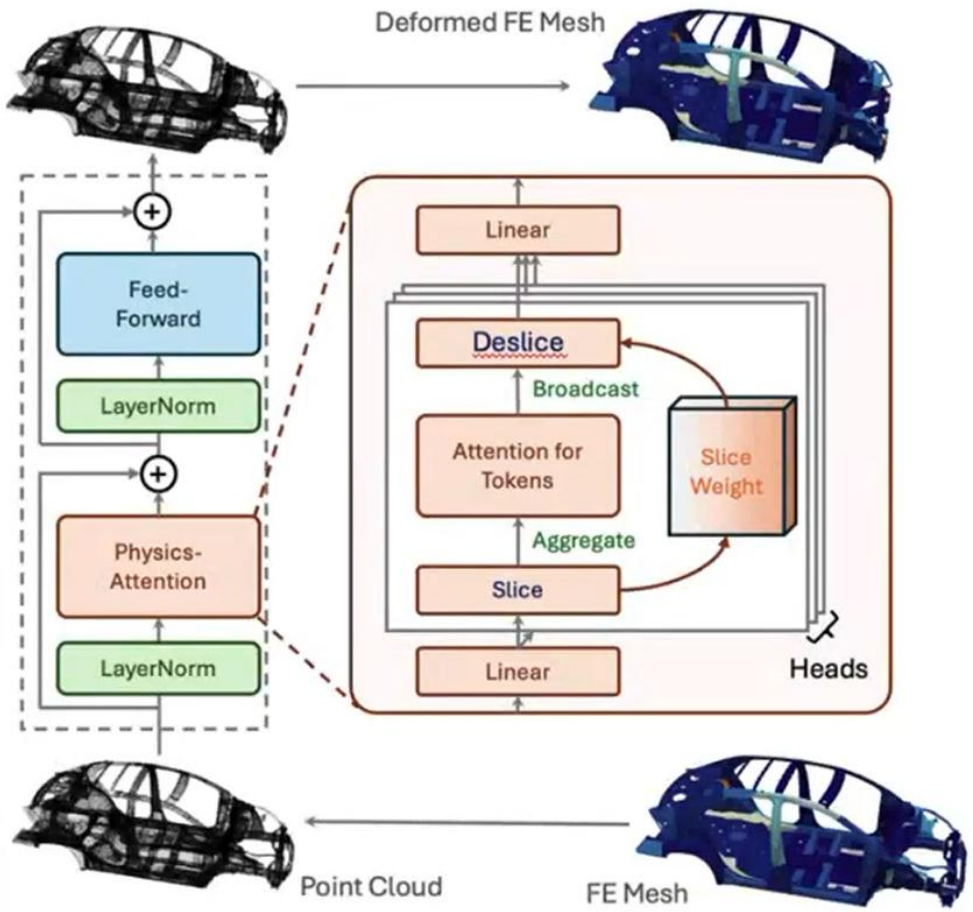
(b) Attention Map Visualization

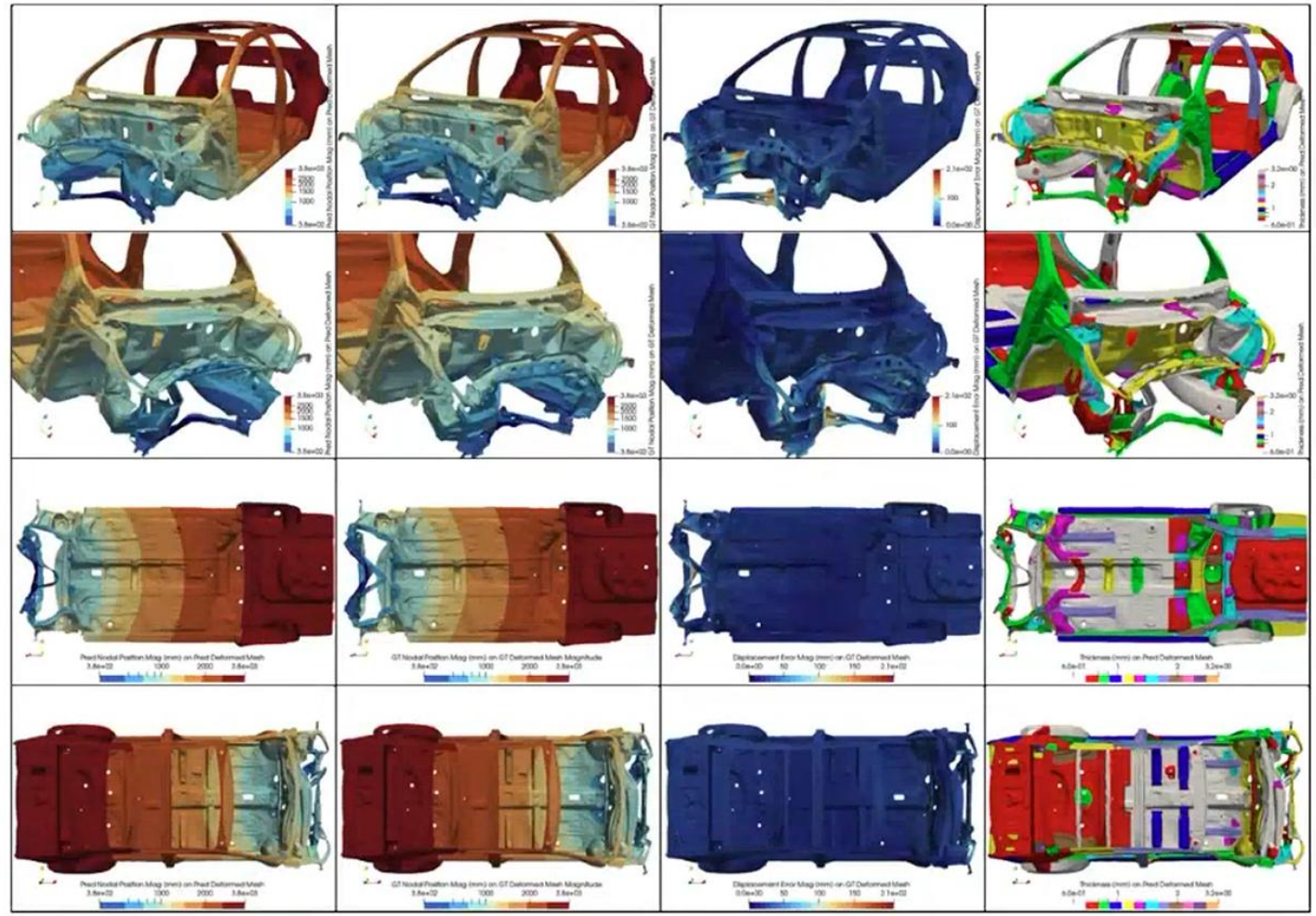


Models	OOD Reynolds		OOD Angles	
	$C_L \downarrow$	$\rho_L \uparrow$	$C_L \downarrow$	$\rho_L \uparrow$
Simple MLP	0.6205	0.9578	0.4128	0.9572
GraphSAGE (2017)	0.4333	0.9707	<u>0.2538</u>	0.9894
PointNet (2017)	0.3836	0.9806	0.4425	0.9784
Graph U-Net (2019)	0.4664	0.9645	0.3756	0.9816
MeshGraphNet (2021)	1.7718	0.7631	0.6525	0.8927
GNO (2020a)	0.4408	<u>0.9878</u>	0.3038	0.9884
Galerkin (2021)	0.4615	0.9826	0.3814	0.9821
GNOT (2023)	<u>0.3268</u>	0.9865	0.3497	0.9868
GINO (2023a)	0.4180	0.9645	0.2583	<u>0.9923</u>
Transolver (Ours)	**0.2996**	**0.9896**	**0.1500**	**0.9950**



NVIDIA PhysicsNeMo





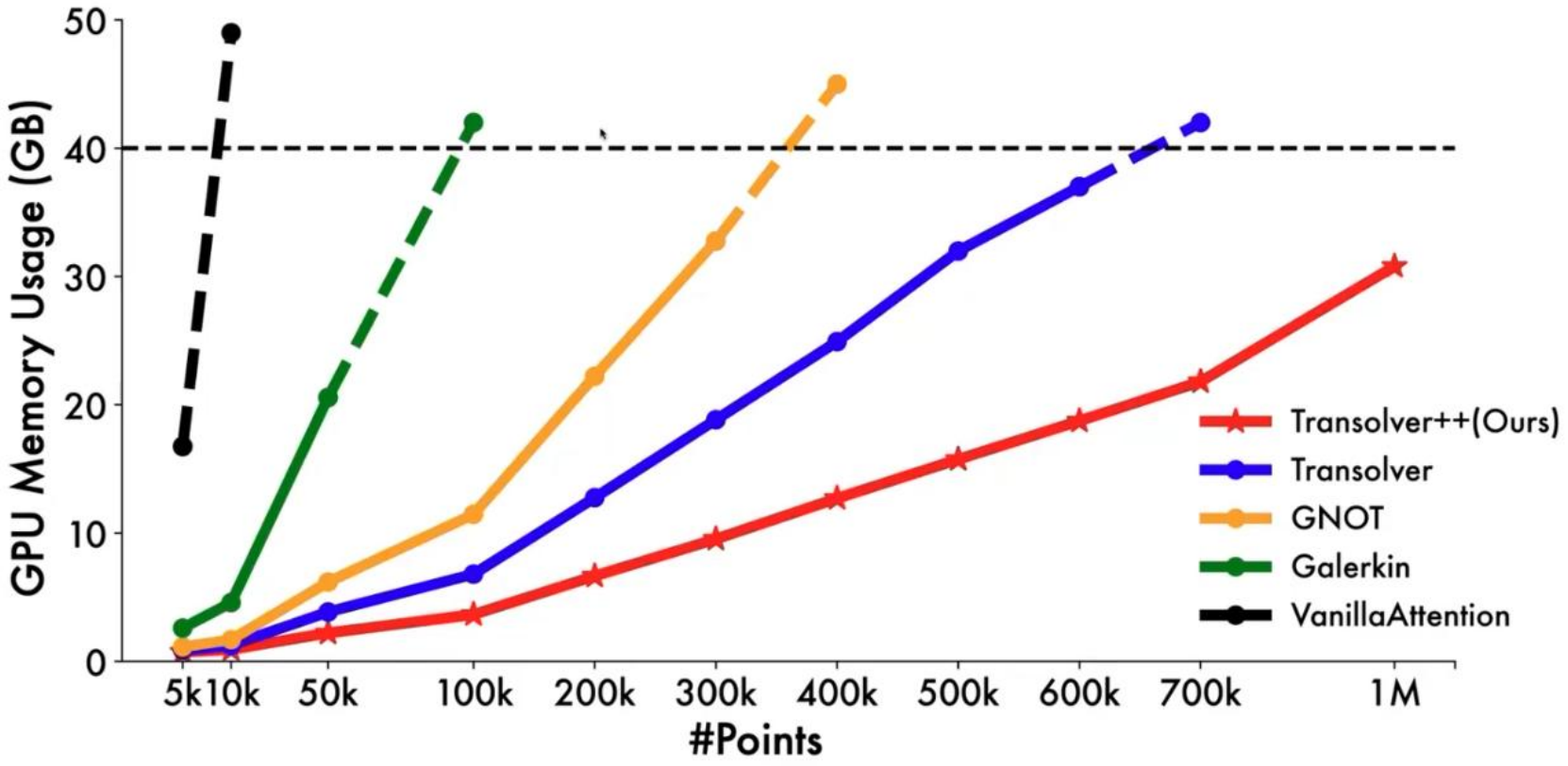
Nabian et al., Automotive Crash Dynamics Modeling Accelerated with Machine Learning, arXiv 2025

Transolver++: An Accurate Neural Solver for PDEs on Million-Scale Geometries

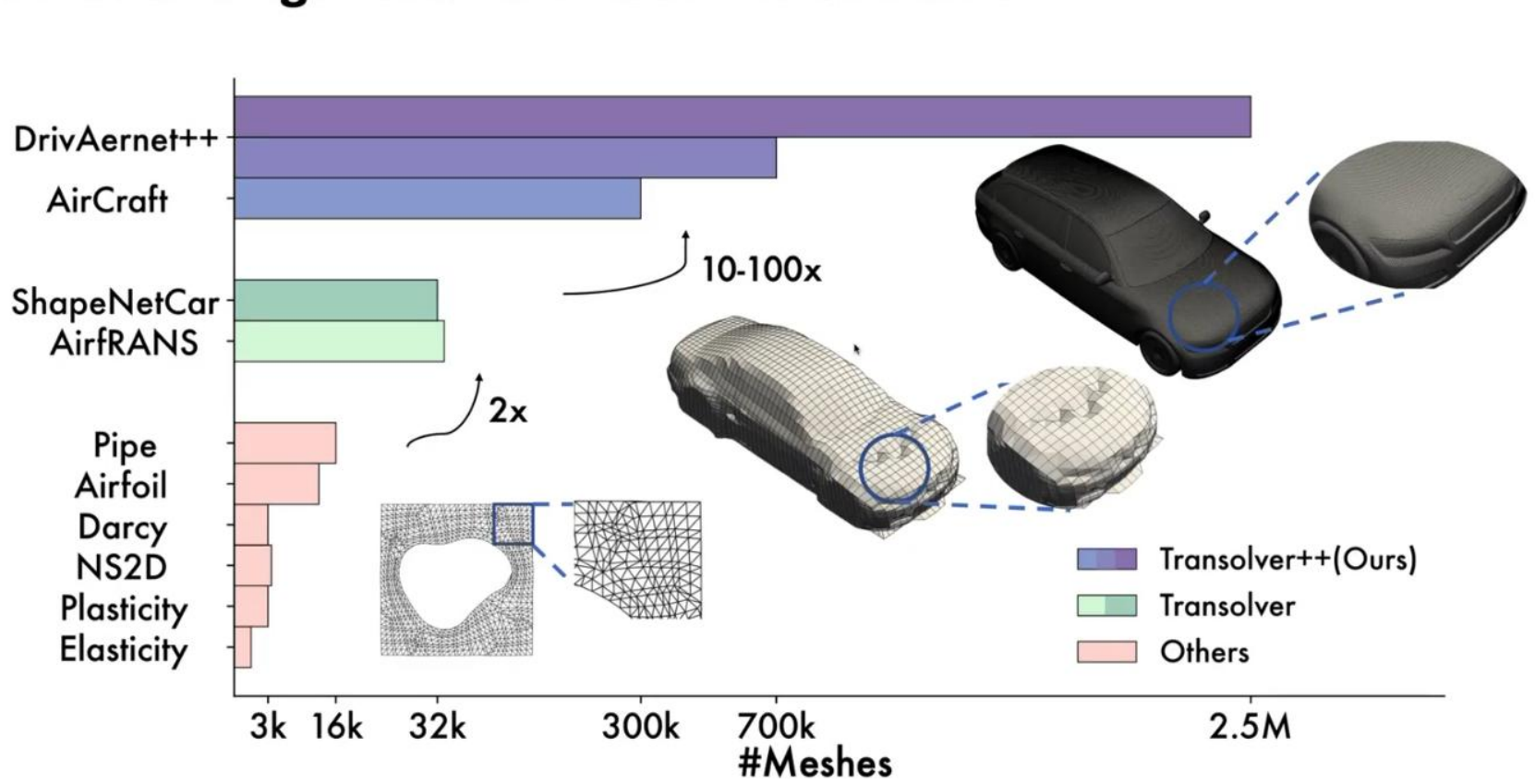
Huakun Luo [*1] Haixu Wu [*1] Hang Zhou [1] Lanxiang Xing [1] Yichen Di [1] Jianmin Wang [1] Mingsheng Long [1]

面向大型工业化的拓展：Transolver++、Transolver3

百万级几何网格数据所需要的显存



超大型几何网络



Large Geometrics In real-world applications

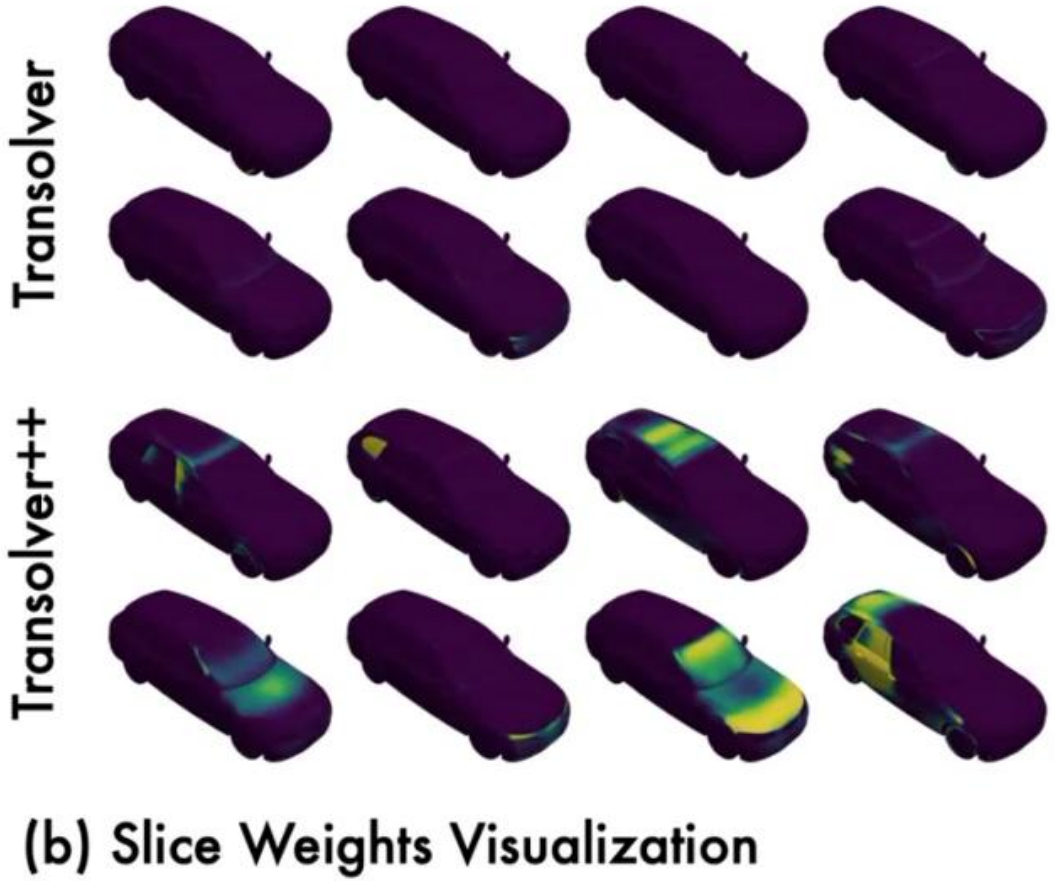
1. More complex geometrics with plenty of details

2. Deep models are expected to be Scalable

3. Models are expected to be more accurate

Challenges within Transolver in Million-Scale Geometries

（1）Transolver要从庞大的候选点找到相应匹配的点



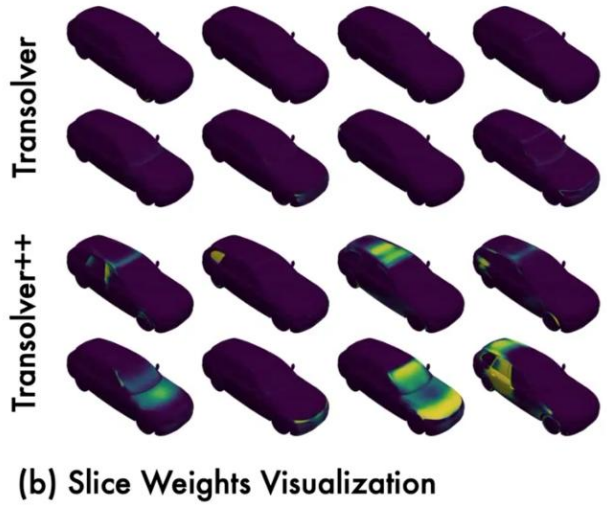
Degenerate in large-scale geometries

Improved physics learning

增加更“丰富”的物理学习

Challenges within Transolver in Million-Scale Geometries

（2）压缩Slice对每个网格点分配权重



增加更“丰富”的物理学习

百万级网格点

$N = 10^6,\ \mathbf{X} \in \mathbb{R}^{N \times C}$

↓

逐点计算 Slice 权重

显存瓶颈

高维特征矩阵 + $N \times M$ 权重矩阵 + 训练激活

↓

加权聚合

少量 Slice 状态

$\mathbf{S} \in \mathbb{R}^{M \times C}$，其中 $M \ll N$

$$\text{Slice weights: } \mathbf{w} = \text{Softmax}\left(\text{Linear}(\mathbf{x})/\tau_0\right)$$

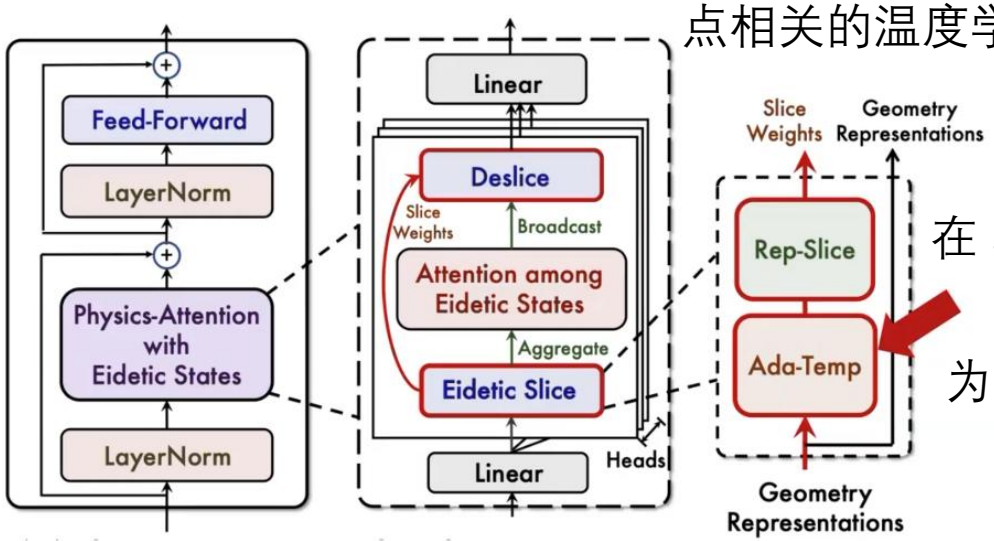
$$\text{Physical states: } \{\mathbf{s}_j\}_{j=1}^{M} = \left\{ \frac{\sum_{i=1}^{N} \mathbf{w}_{ij}\mathbf{x}_i}{\sum_{i=1}^{N} \mathbf{w}_{ij}} \right\}_{j=1}^{M}$$

一百万个网格点会消耗 2 GB 显存

$$\text{Memory} = N \times C \times b,$$

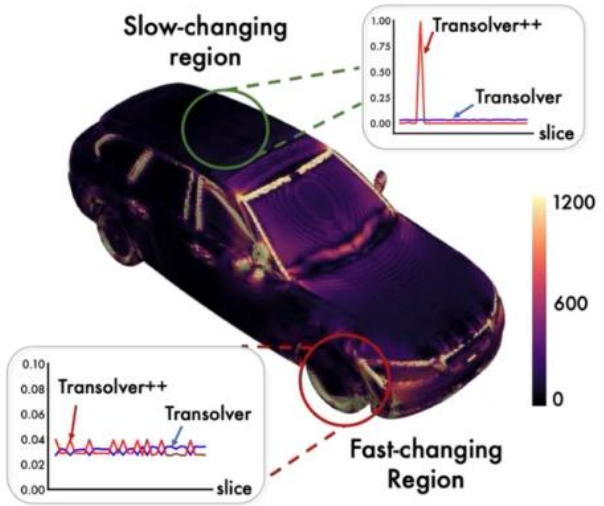
Upgrade1： Architectural Design

点相关的温度学习不同位置的分配灵活性



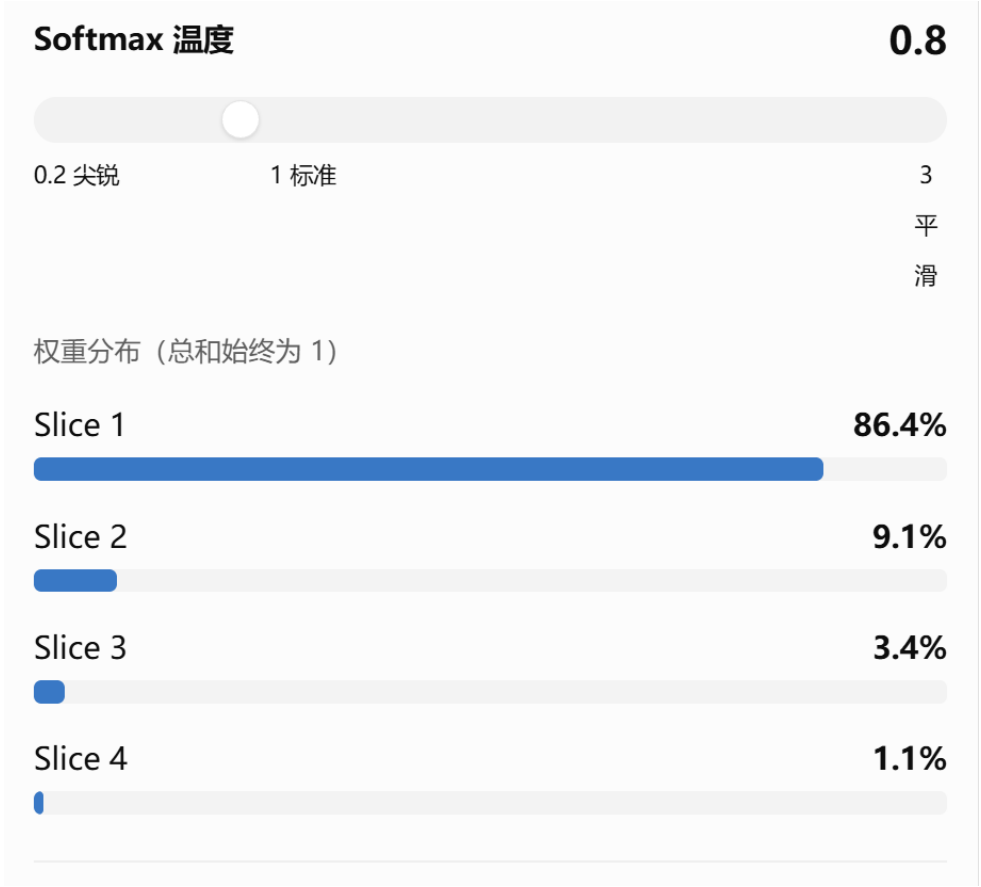
在 Softmax 中加入随机扰动学习每个点的不确定性

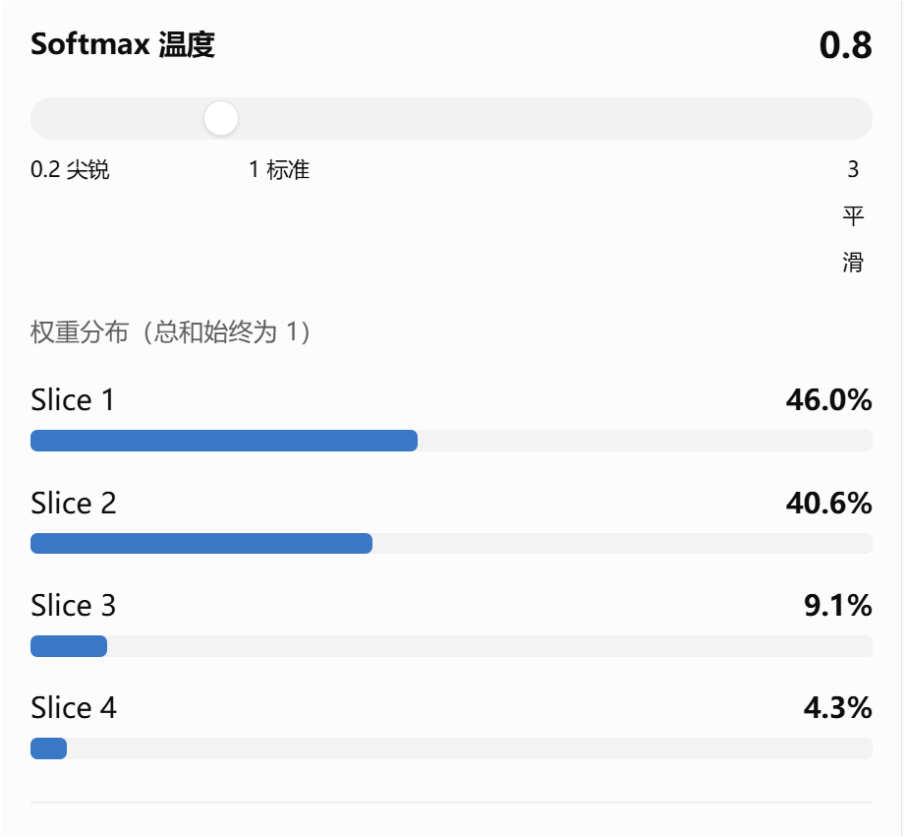
为每个点预τ，自适应调节分配的软硬程度



$$\text{Ada-Temp: } \tau = \{\tau_i\}_{i=1}^{N} = \{\tau_0 + \text{Linear}(\mathbf{x}_i)\}_{i=1}^{N},$$

$$\text{Rep-Slice}(\mathbf{x}, \tau) = \text{Softmax}\left(\frac{\text{Linear}(\mathbf{x}) - \log(-\log \epsilon)}{\tau}\right), \tag{4}$$



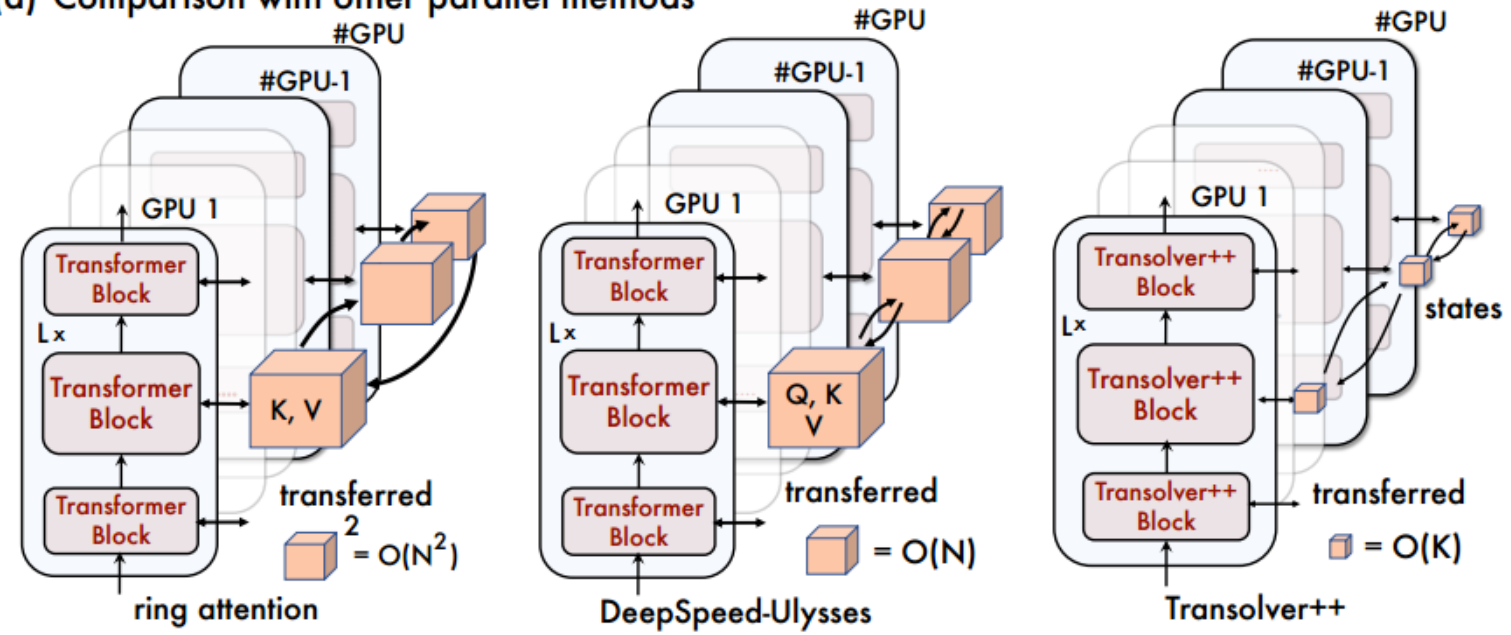


它让不同网格点可以采用不同的分配温度

Upgrade2： 多GPU并行计算:利用可加性

Additivity of physical states: $$\mathbf{s}_j = \frac{\sum_{i=1}^{N_1} \mathbf{w}_{ij}^{(1)} \mathbf{x}_i^{(1)} \oplus \cdots \oplus \sum_{i=1}^{N_{\#\text{gpu}}} \mathbf{w}_{ij}^{(\#\text{gpu})} \mathbf{x}_i^{(\#\text{gpu})}}{\sum_{i=1}^{N_1} \mathbf{w}_{ij}^{(1)} \oplus \cdots \oplus \sum_{i=1}^{N_{\#\text{gpu}}} \mathbf{w}_{ij}^{(\#\text{gpu})}}$$

(a) Comparison with other parallel methods



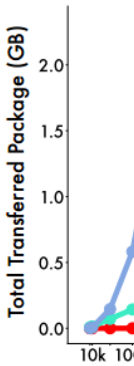
Equivalent result

Accumulate multi-GPU results

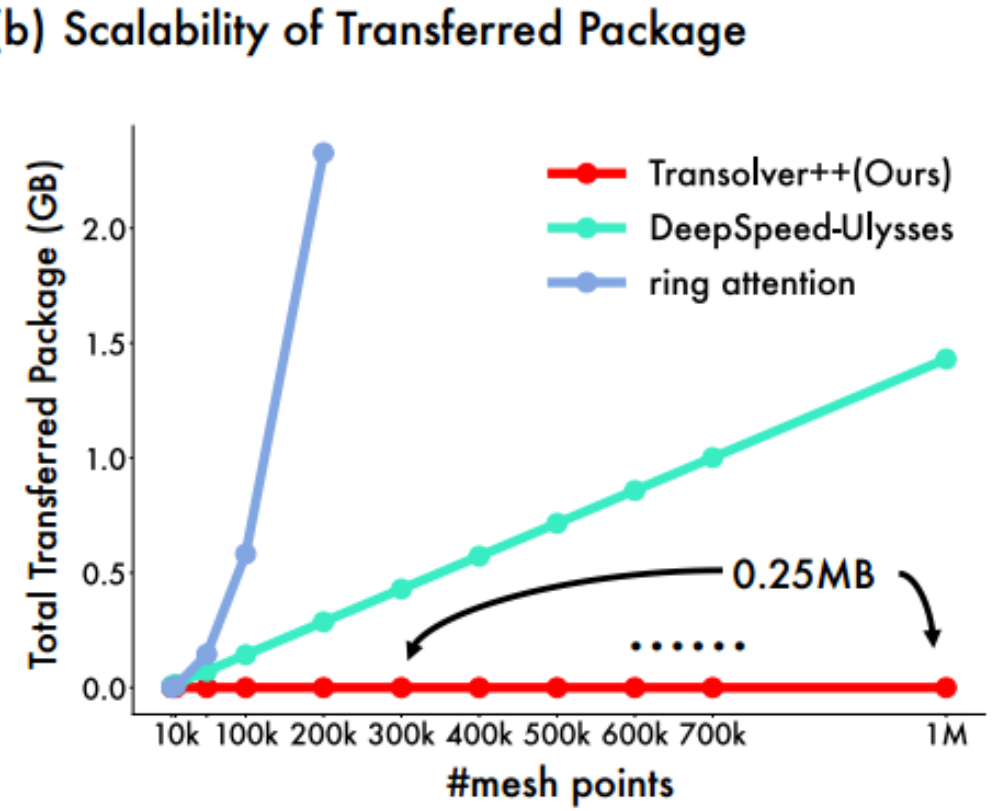
Compute physical states in each GPU

Split the mesh into multiple GPUs

(b) Scalabilit



Upgrade2：多GPU并行计算——多GPU之间通信数据Package



Equivalent result

↑

Accumulate
multi-GPU results

↑

Compute physical states
in each GPU

↑

Split the mesh into
multiple GPUs

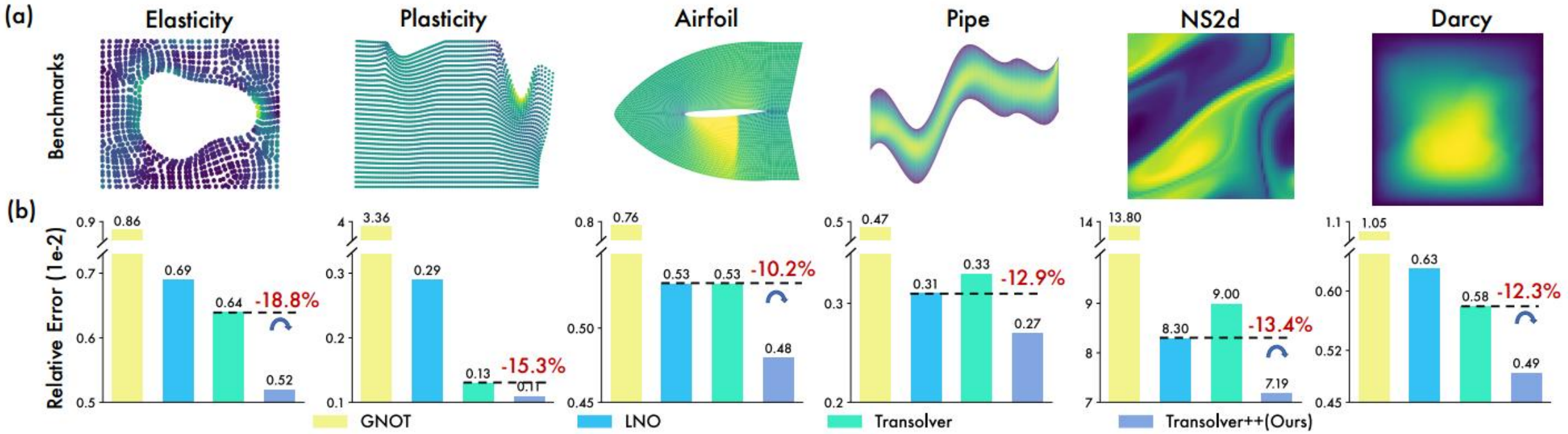
Additivity of physical states:

$$\mathbf{s}_j = \frac{\sum_{i=1}^{N_1} \mathbf{w}_{ij}^{(1)} \mathbf{x}_i^{(1)} \oplus \cdots \oplus \sum_{i=1}^{N_{\#\text{gpu}}} \mathbf{w}_{ij}^{(\#\text{gpu})} \mathbf{x}_i^{(\#\text{gpu})}}{\sum_{i=1}^{N_1} \mathbf{w}_{ij}^{(1)} \oplus \cdots \oplus \sum_{i=1}^{N_{\#\text{gpu}}} \mathbf{w}_{ij}^{(\#\text{gpu})}}$$

Transolveer++： Experiment

数据集：
标准PDE求解+
大Meshes工业场景

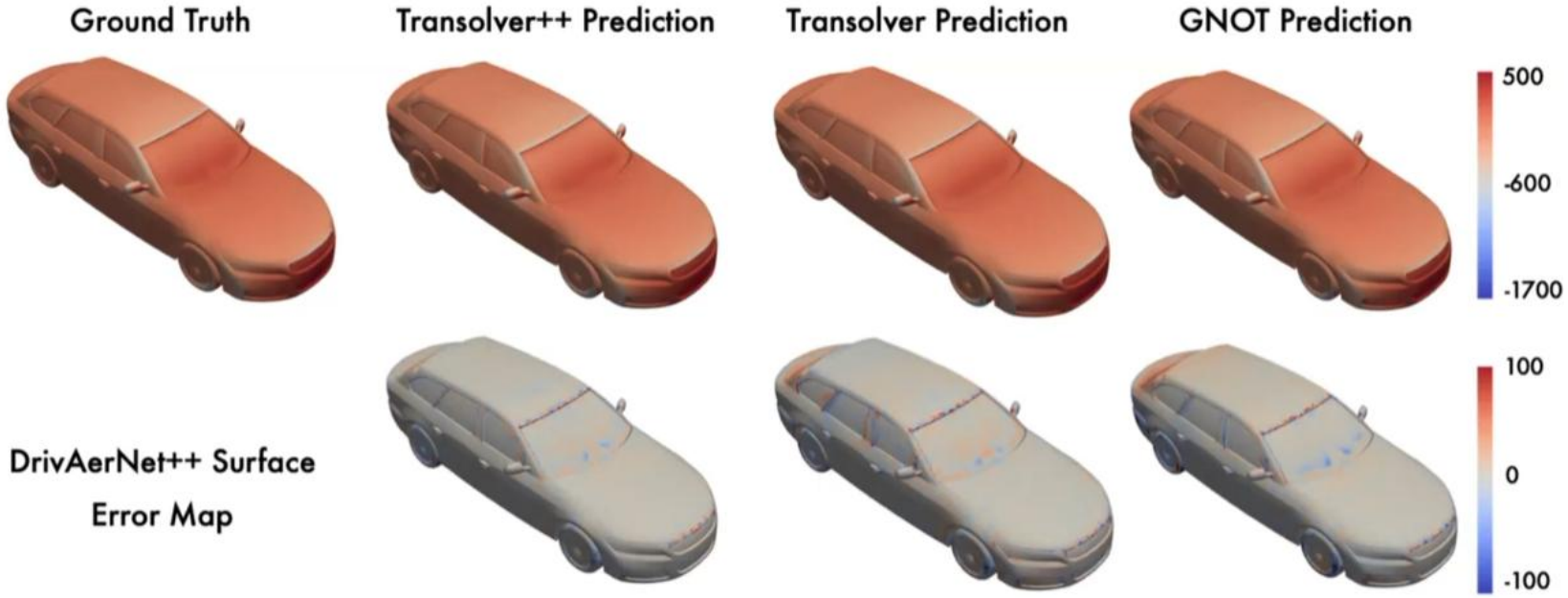
Type	Benchmarks	Geo Type	#Mesh
Standard Benchmarks	NS2D	Structure	4,096
	Pipe	Structure	16,641
	Darcy	Structure	7,225
	Airfoil	Structure	11,271
	Plasticity	Structure	3,131
	Elasticity	unstructure	972
Industrial Applications	AirCraft	unstructure	~300K
	DrivAerNet++		~2.5M



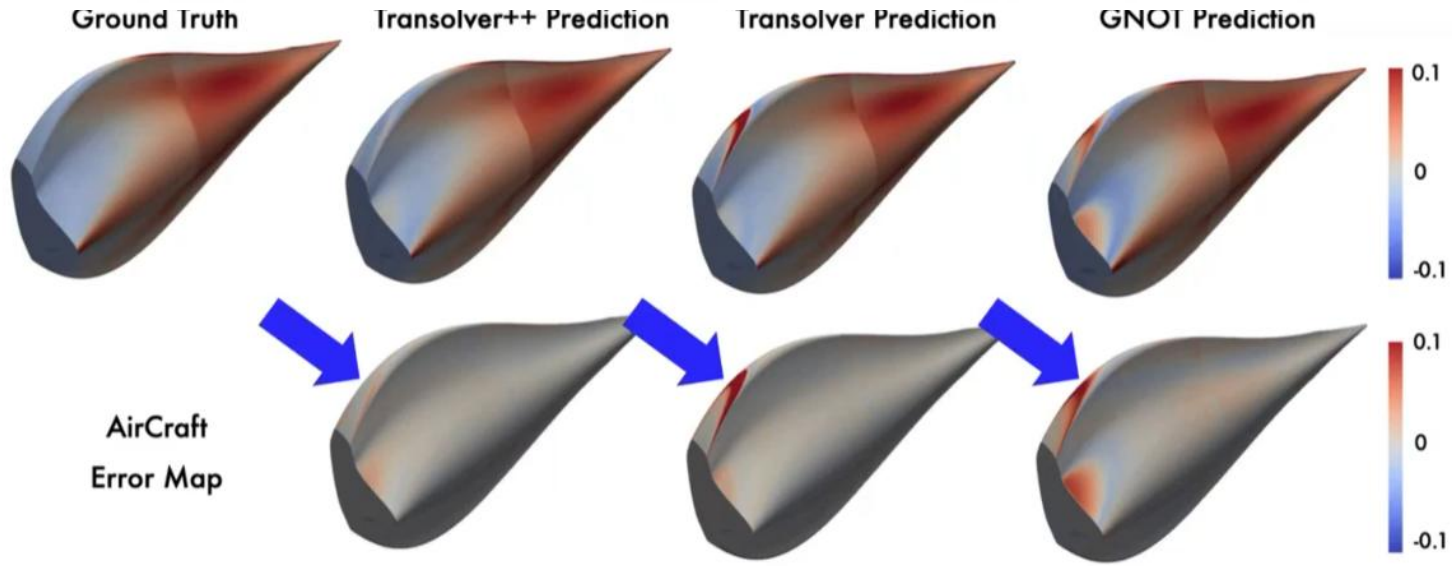
Transolveer++： Experiment

Table 2. Comparison on large geometries benchmarks. Relative L2 of the surrounding area (*Volume*) and surface (*Surf*) physics as well as drag and lift coefficient (C_D, C_L) is recorded, along with their coefficient of determination R_D^2 and R_L^2. The closer R^2 is to 1, the better.

Model	DrivAerNet++ Full		DrivAerNet++ Surf			AirCraft		
	Volume ↓	Surf ↓	C_D ↓	R_L^2 ↑	Surf ↓	C_L ↓	R_L^2 ↑	Surf ↓
GraphSAGE (2017)	0.328	0.284	0.282	0.859	0.294	0.040	0.988	0.109
PointNet (20 跳转页面位置:9	0.285	0.478	0.301	0.831	0.237	0.095	0.982	0.169
Graph U-Net* (2019)	0.241	0.260	0.272	0.876	0.193	0.063	0.953	0.161
MeshGraphNet* (2021)	0.529	0.422	0.260	0.870	0.209	0.038	0.993	0.113
GNO* (2020a)	0.510	0.664	0.252	0.882	0.196	0.031	0.991	0.129
Galerkin* (2021)	0.234	0.274	0.267	0.792	0.235	0.069	0.879	0.118
Geo-FNO* (2022)	0.718	0.892	0.288	0.831	0.291	0.243	0.903	0.395
GINO (2023a)	0.586	0.638	0.323	0.725	0.220	0.047	0.983	0.133
GNOT* (2023)	0.174	0.171	0.158	0.901	0.167	0.033	0.991	0.093
LNO* (2024)	0.180	0.203	0.208	0.855	0.195	0.091	0.992	0.137
3D-GeoCA* (2024)	0.389	0.224	0.205	0.883	0.175	0.022	0.993	0.097
Transolver* (2024)	0.173	0.167	0.061	0.931	0.145	0.037	0.994	0.092
Transolver++ (Ours)	**0.154**	**0.146**	**0.036**	**0.997**	**0.110**	**0.014**	**0.999**	**0.064**
Relative Promotion	11.0%	12.6%	41.0%	-	24.1%	36.3%	-	30.4%



Transolver++ achieves over 20% error reduction than other models.

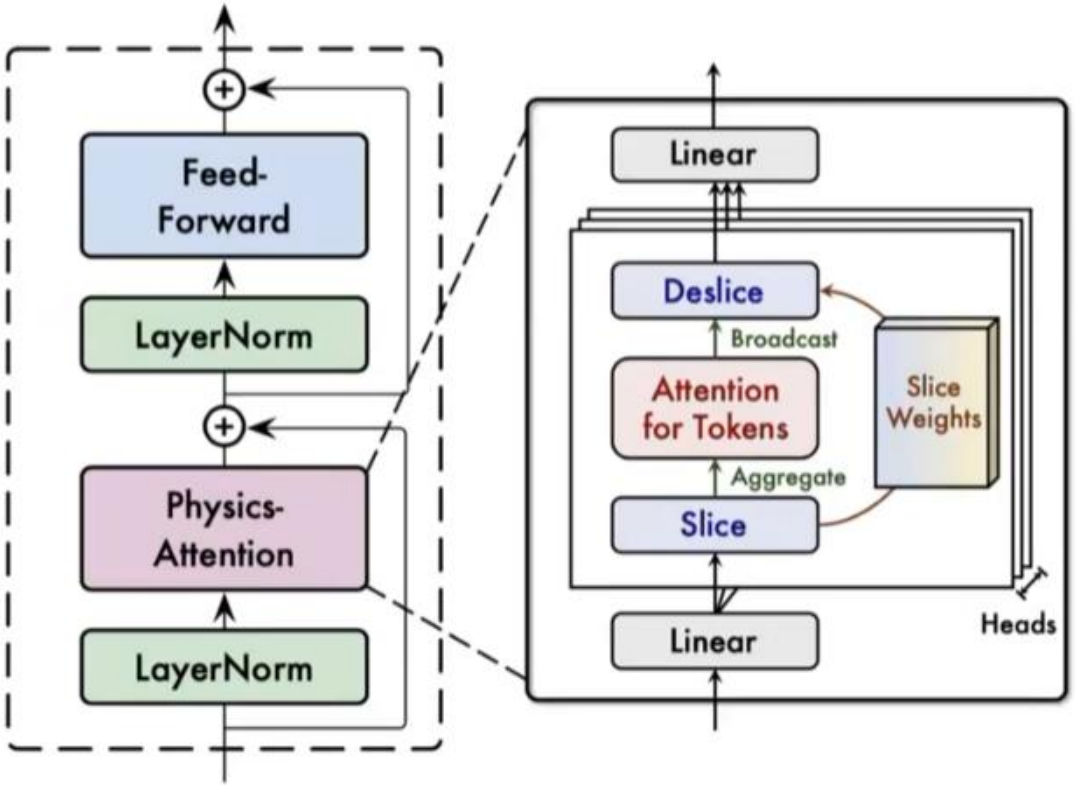


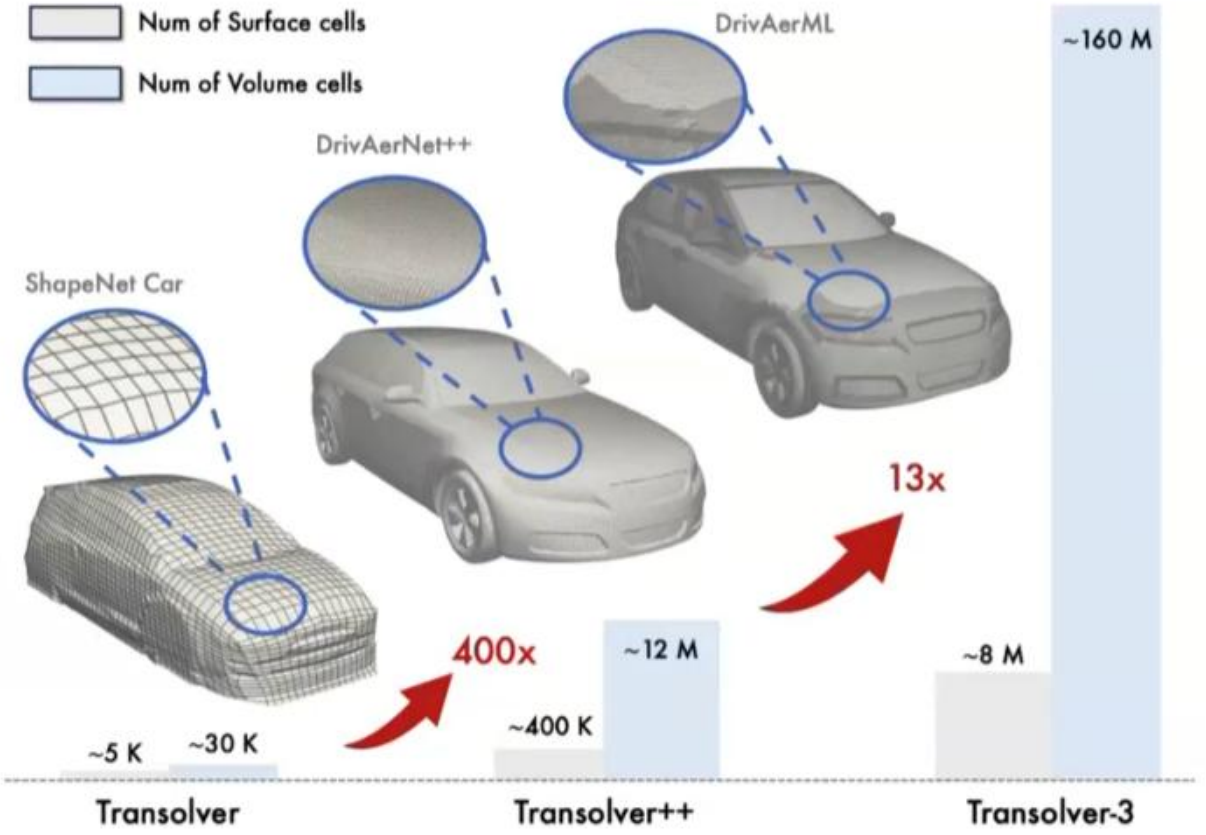
Transolver-3: Scaling Up Transformer Solvers to Industrial-Scale Geometries

Hang Zhou [1] Haixu Wu [1] Haonan Shangguan [1] Yuezhou Ma [1] Huikun Weng [1] Jianmin Wang [1]
Mingsheng Long [1]

面向工业规模的超大 meshes： Transolver3
支持仅单个GPU运行

Scale to Over 100-Million-Cell Geometries





原始Transolver的计算复杂度

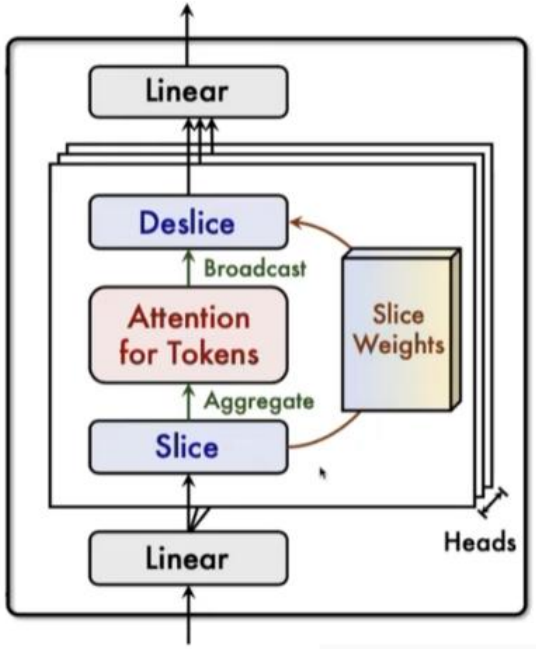


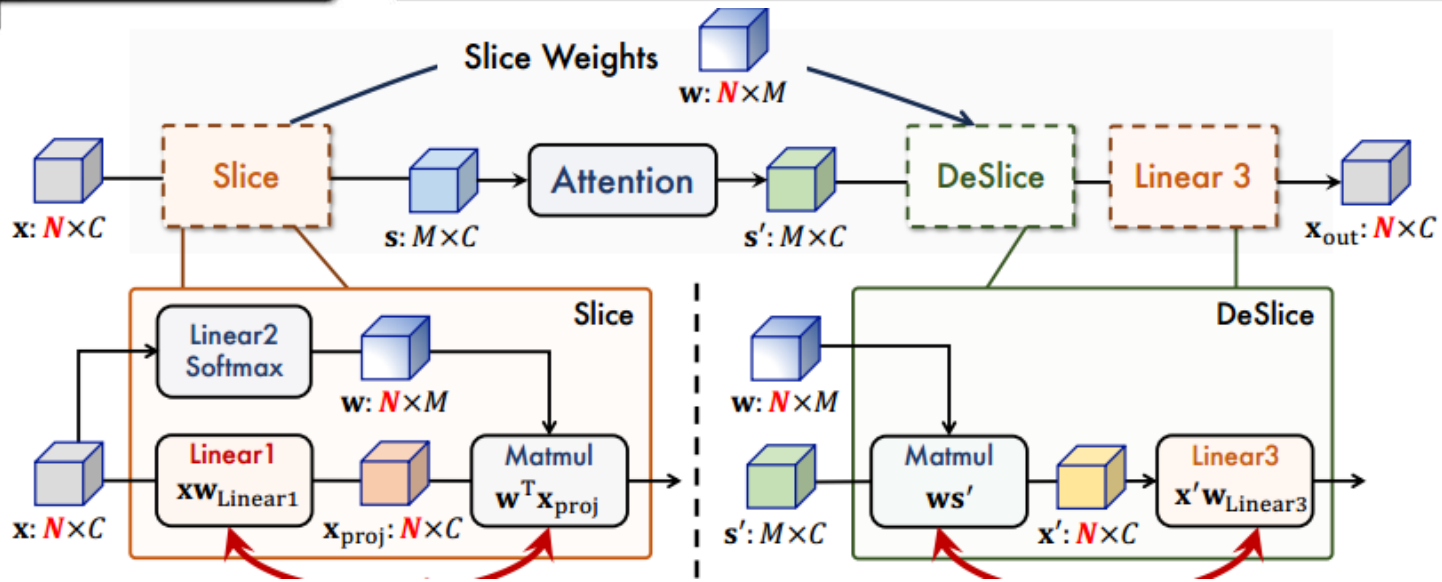
Table 1. Complexity Analysis of Original Physics-Attention.

Operation	Time Complexity	Space Complexity
$\text{Linear1}(\mathbf{x})$	$O(NC^2)$	$O(NC)$
$\text{Softmax}(\text{Linear2}(\mathbf{x}))$	$O(NCM)$	$O(NM)$
$(\mathbf{w}\mathbf{d}^{-1})^\top \mathbf{x}_{\text{proj}}$	$O(NMC)$	$O(MC)$
$\text{Attention}(\mathbf{s})$	$O(M^2C)$	$O(M^2+MC)$
$\mathbf{w}\mathbf{s}'$	$O(NMC)$	$O(NC)$
$\text{Linear3}(\mathbf{w}\mathbf{s}')$	$O(NC^2)$	$O(NC)$
N-Related Terms	5	4

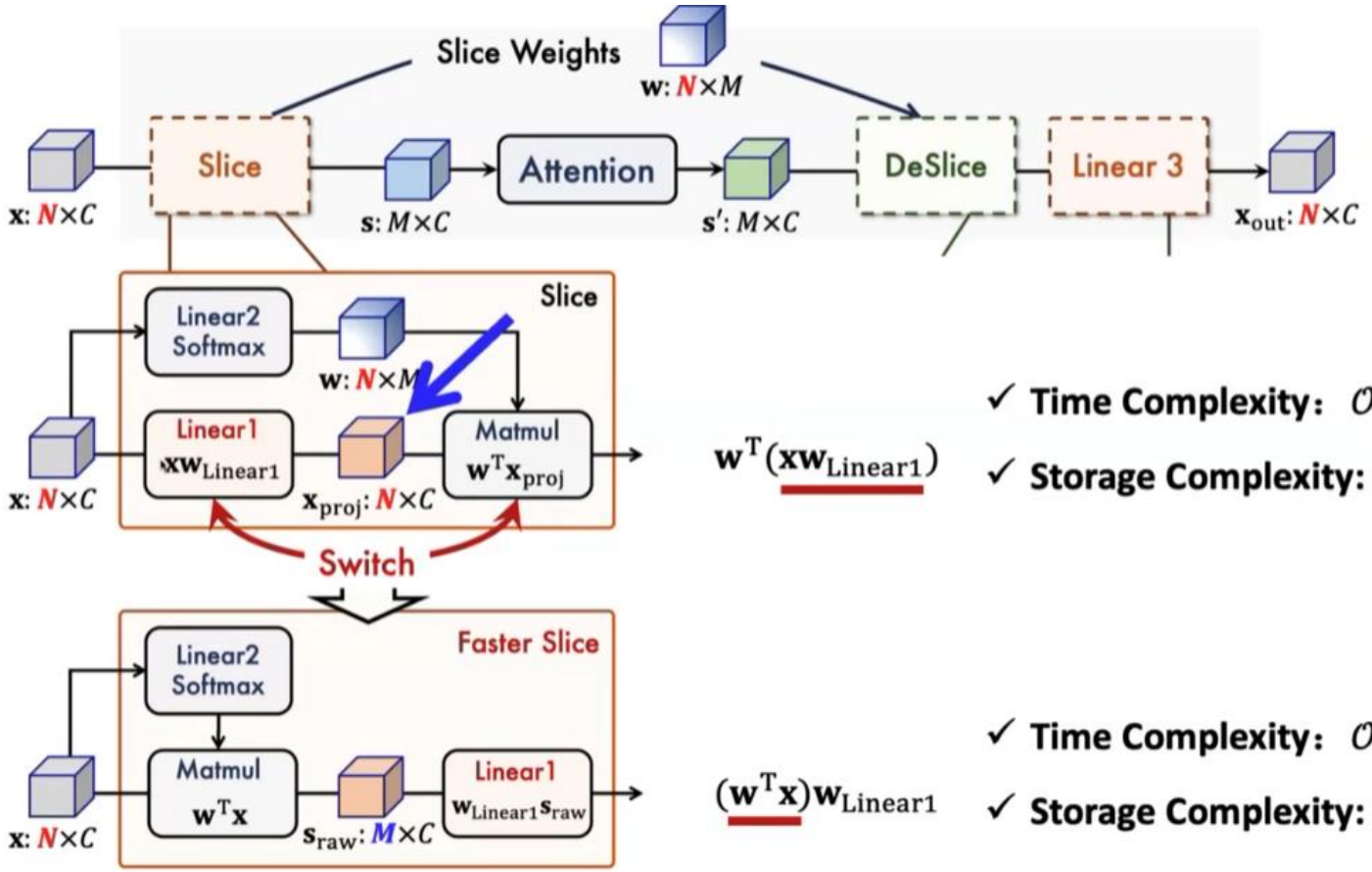
Slice

Attn

Deslice



Upgrade1： Architecture Design



$\mathbf{w}^{\mathrm{T}}(\mathbf{x}\mathbf{w}_{\mathrm{Linear1}})$

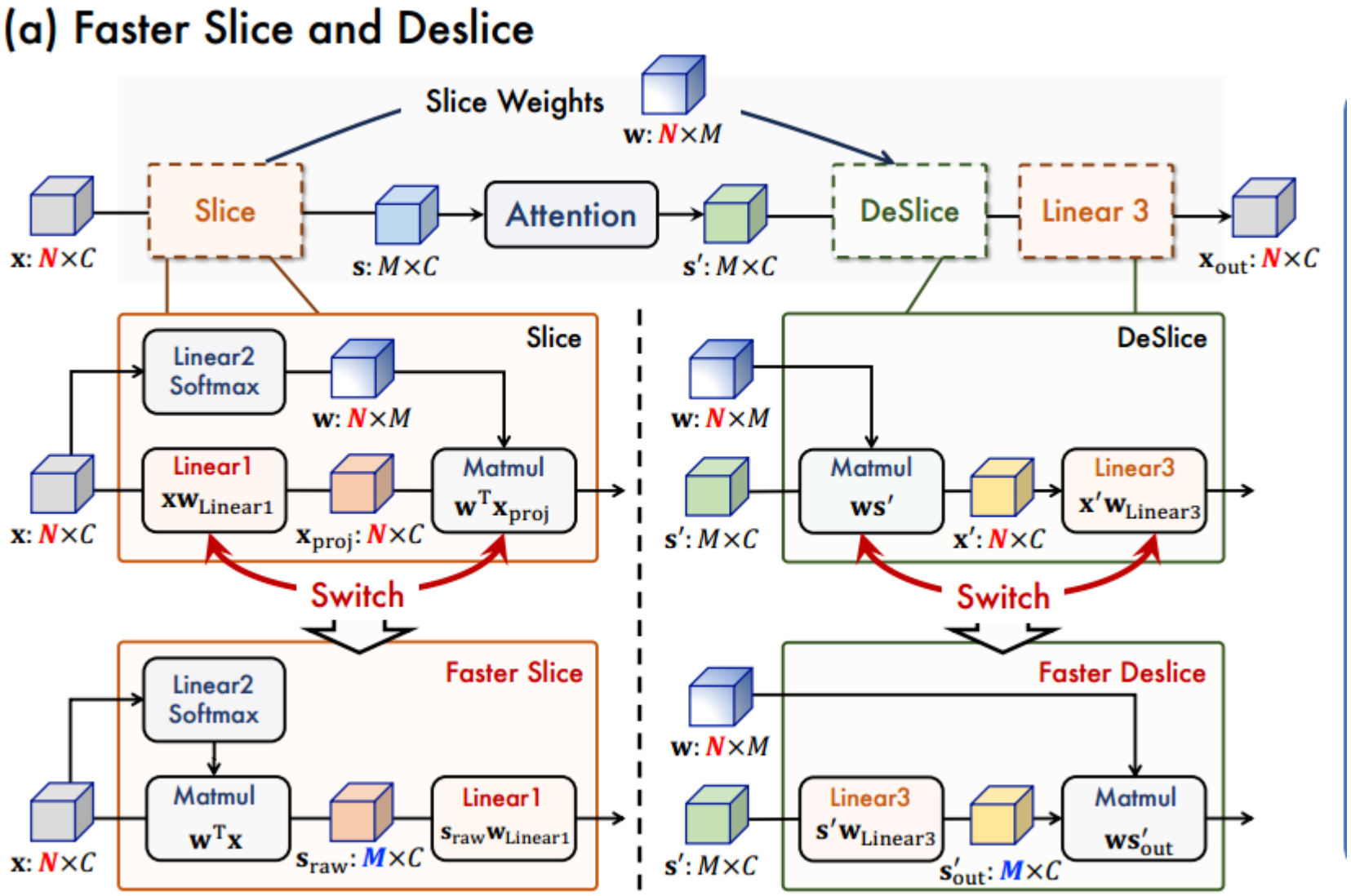
- ✓ **Time Complexity：** $\mathcal{O}(\boldsymbol{NC^2 + NMC})$
- ✓ **Storage Complexity:** $\mathcal{O}(\boldsymbol{NM + NC})$

$(\mathbf{w}^{\mathrm{T}}\mathbf{x})\mathbf{w}_{\mathrm{Linear1}}$

- ✓ **Time Complexity：** $\mathcal{O}(\boldsymbol{MC^2 + NMC})$
- ✓ **Storage Complexity:** $\mathcal{O}(\boldsymbol{NM + MC})$

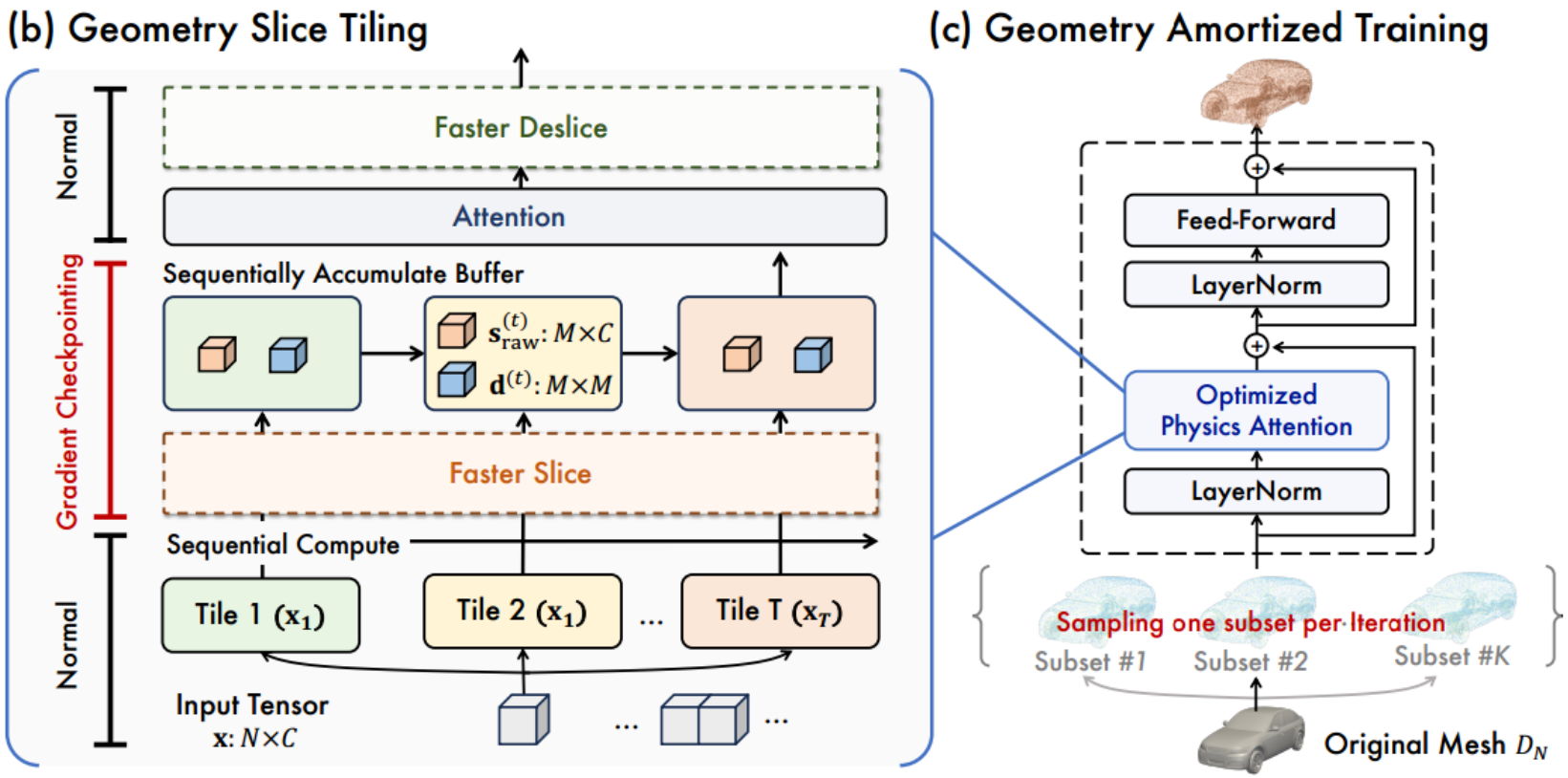
利用矩阵乘法结合律;
Linear1 处理N行特征->M行特征

Upgrade1： Architecture Design



Upgrade1：Architecture Design

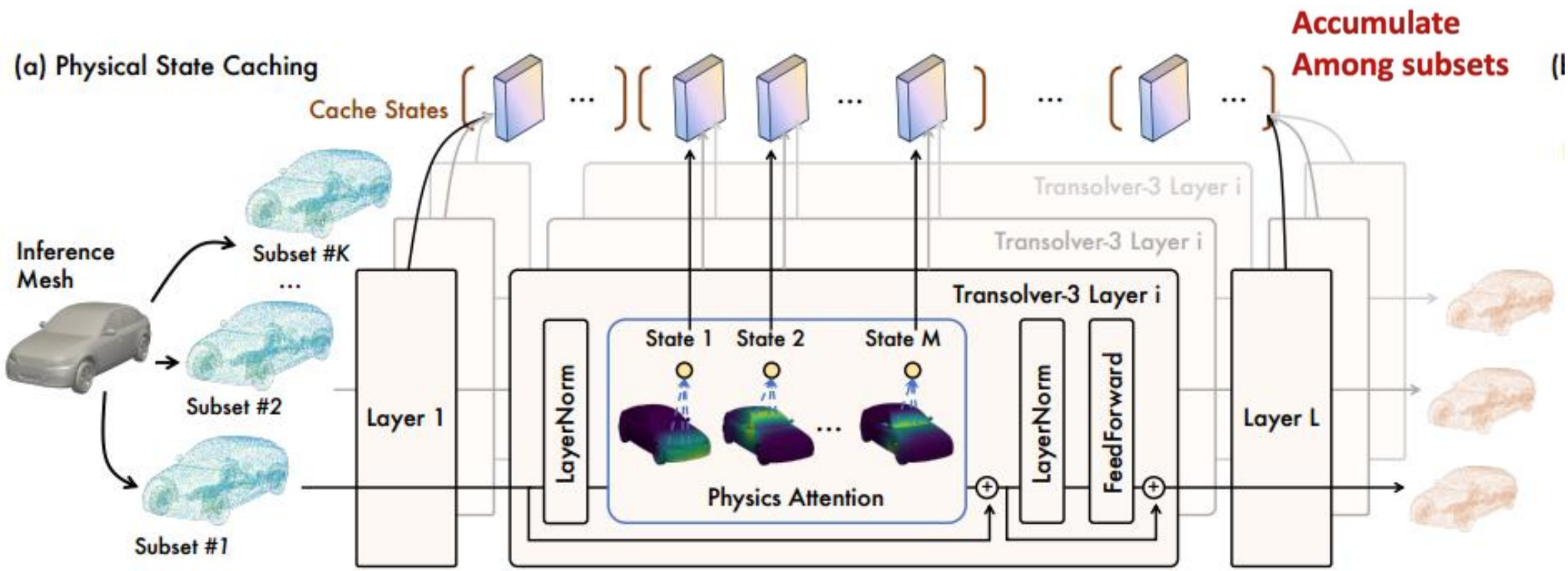
大网格训练问题：peak memory usage、training cost



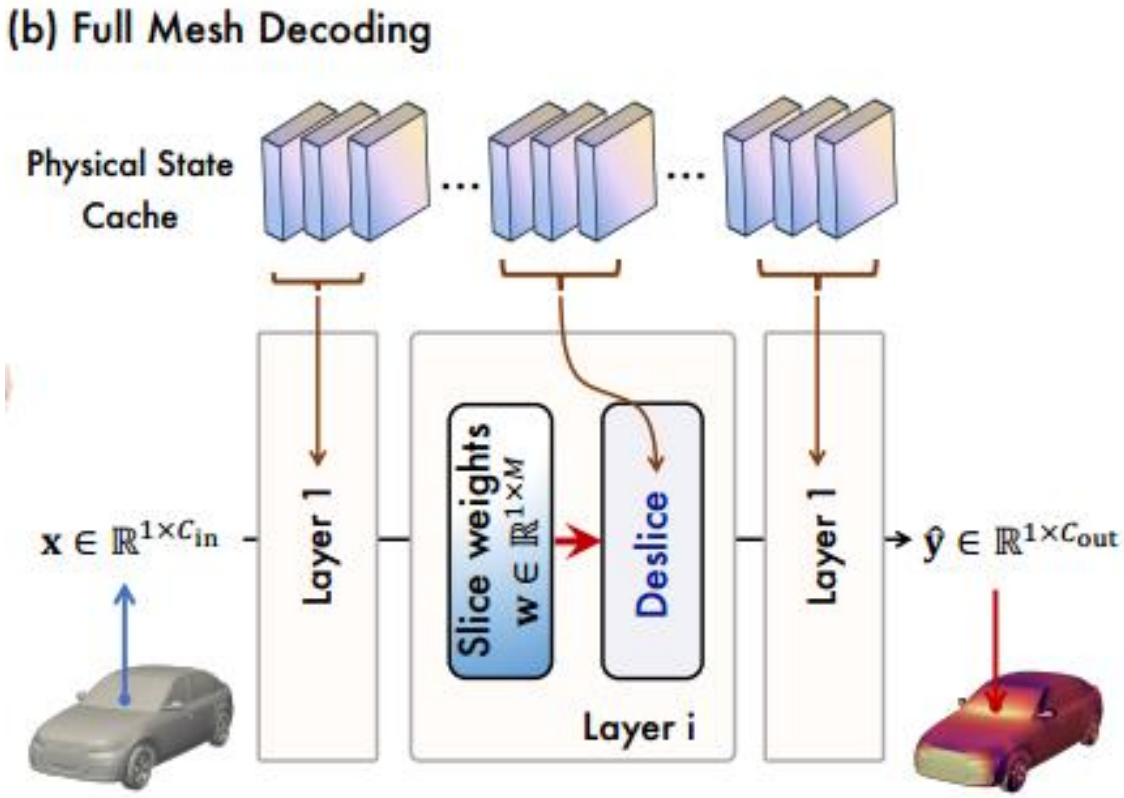
大网格拆成多个Tile，分块计算slice：
顺序累积、设立梯度检查点
降低训练时的峰值显存。

每次迭代只从原始网格中采样一个子集进行训练，让多个迭代逐步覆盖整个几何网格

Upgrade2： Inference Scaling Framework



任意坐标点的推理



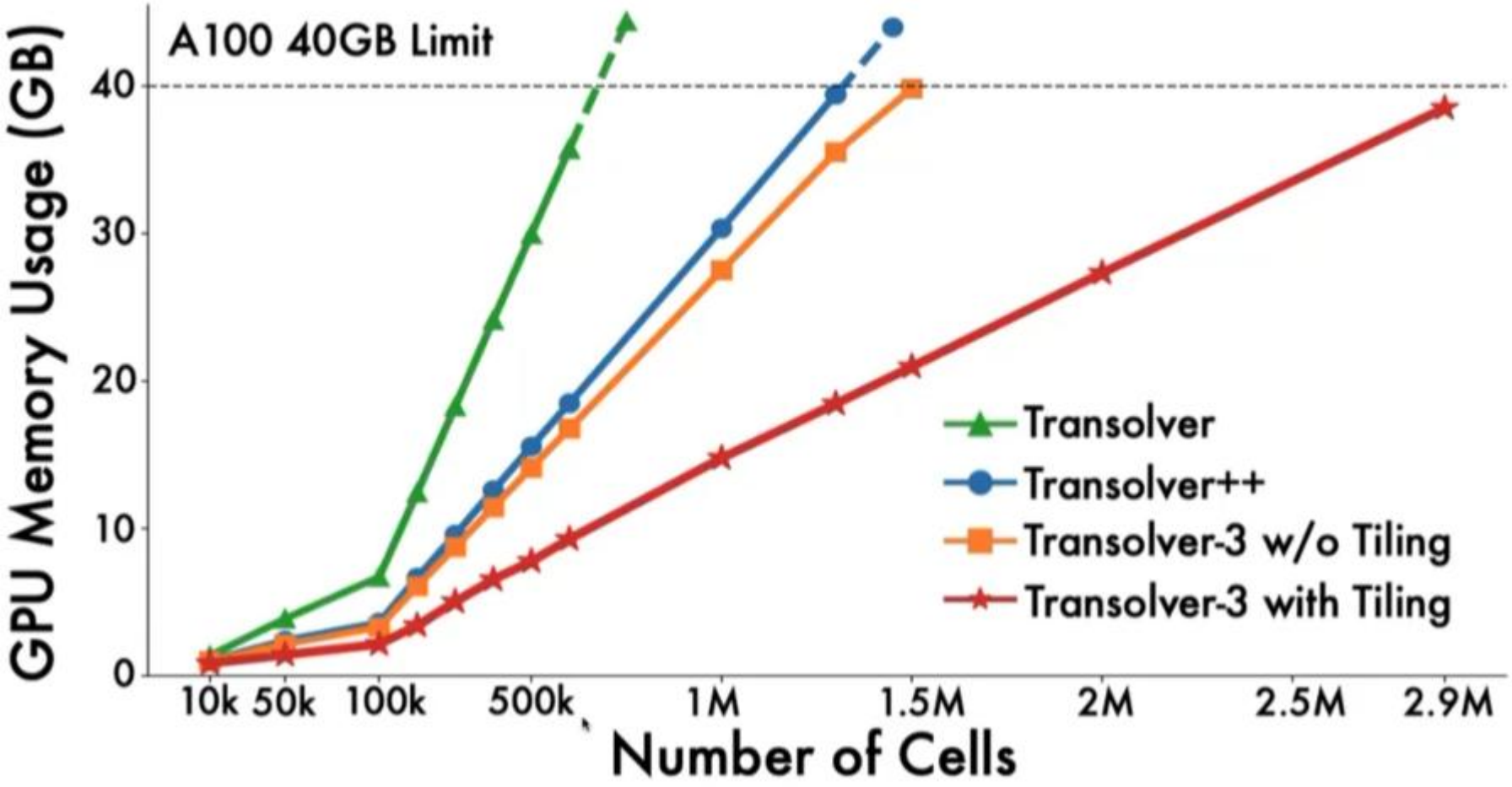
$$\mathbf{w}^{(l)} = \text{Softmax}\left(\text{Linear2}\left(\mathbf{x}^{(l)}\right)\right)$$

$$\mathbf{x}_{\text{out}}^{(l)} = \mathbf{w}^{(l)}\mathbf{s}_{\text{out}}^{\prime(l)}$$

Cached physical states

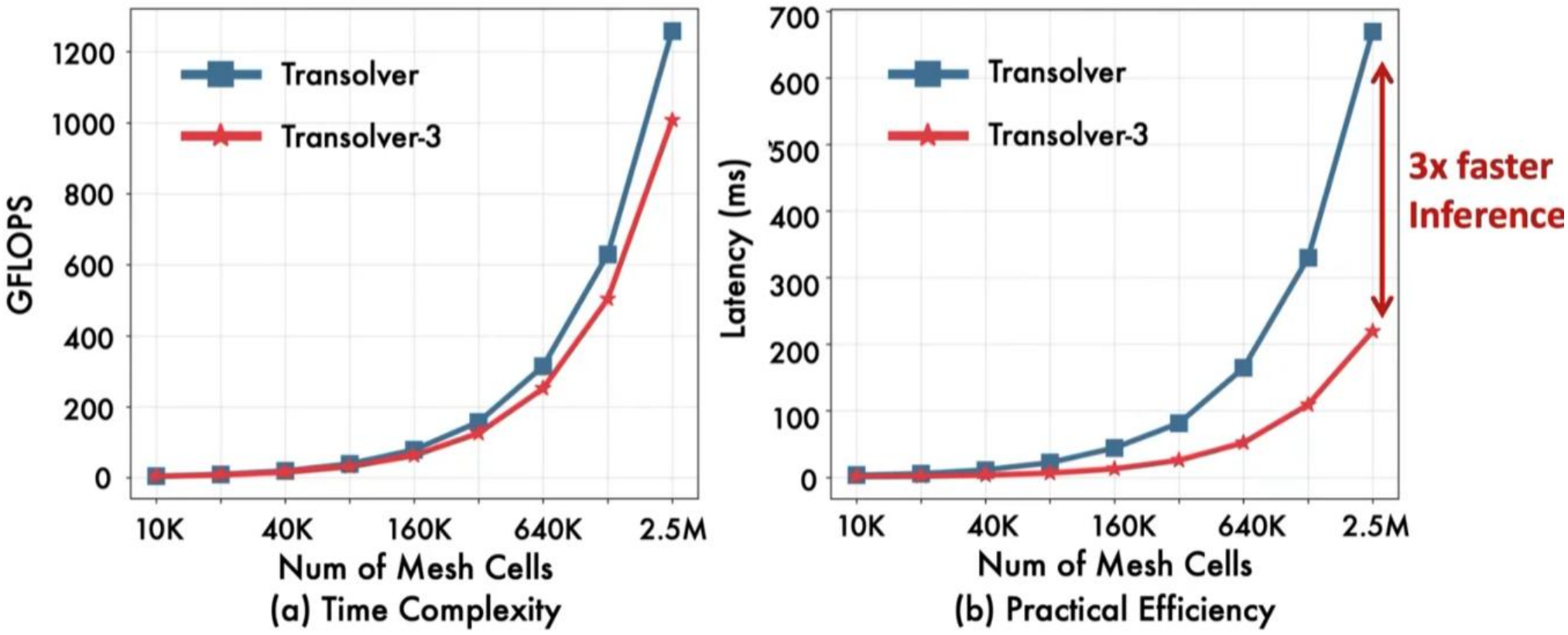
Newly estimated slice weights

Efficiency Analysis



比Transolver 大5倍，比Transolver++ 大2倍

Efficiency Analysis



GFLOPs，表示十亿次浮点运算

Transolver-3 不仅计算量更低，而且实际推理速度也更快。

Experiment Results

BENCHMARKS	GEO TYPE	#MESH	SIZE
NASA-CRM	SURFACE	∼400K	∼4GB
AHMEDML	SURFACE VOLUME	∼1M ∼20M	∼8TB
DRIVAERML	SURFACE VOLUME	∼8M ∼160M	∼31TB

Table 4. Relative L2 errors (in %) of surface pressure $\boldsymbol{p}_s$ and skin friction coefficient $\boldsymbol{C}_f$ on the NASA-CRM dataset, and surface pressure $\boldsymbol{p}_s$, volume velocity $\boldsymbol{u}$, wall shear stress $\boldsymbol{\tau}$ and volume pressure $\boldsymbol{p}_v$ on the AhmedML and DrivAerML datasets.

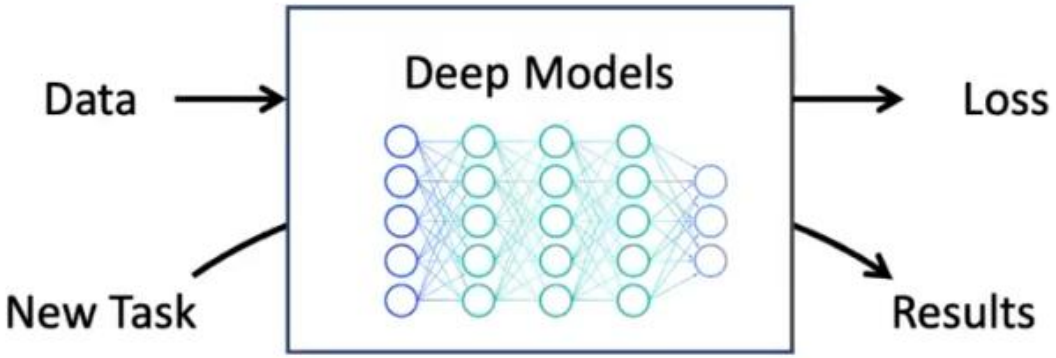
MODELS	NASA-CRM		AHMEDML				DRIVAERML			
	$\boldsymbol{p}_s$	$\boldsymbol{C}_f$	$\boldsymbol{p}_s$	$\boldsymbol{u}$	$\boldsymbol{\tau}$	$\boldsymbol{p}_v$	$\boldsymbol{p}_s$	$\boldsymbol{u}$	$\boldsymbol{\tau}$	$\boldsymbol{p}_v$
GRAPH U-NET*	15.85	15.61	6.46	4.15	7.29	5.18	16.13	17.98	27.84	20.51
GINO*	12.39	11.51	7.90	6.23	8.18	8.80	13.03	40.58	21.71	44.90
MESHGRAPHNET*	14.80	9.81	3.72	3.97	5.01	4.50	14.30	15.01	20.13	21.82
GAOT*	30.38	59.79	8.02	7.43	9.92	10.47	34.00	57.18	61.00	56.90
UPT	12.78	23.78	4.25	2.73	5.80	3.10	7.44	8.74	12.93	10.05
AB-UPT	9.77	<u>6.43</u>	3.97	1.94	5.60	**2.07**	<u>3.82</u>	5.93	7.29	<u>6.08</u>
TRANSOLVER*	9.61	7.04	<u>3.20</u>	1.81	<u>4.85</u>	2.41	4.81	6.78	8.95	7.74
TRANSOLVER++*	<u>9.51</u>	6.95	3.47	<u>1.78</u>	5.06	2.35	4.12	<u>4.70</u>	<u>6.42</u>	6.70
TRANSOLVER-3	**8.71**	**5.85**	**2.96**	**1.60**	**4.81**	<u>2.16</u>	**3.71**	**4.14**	**5.85**	**5.72**

GeoPT: Scaling Physics Simulation via Lifted Geometric Pre-Training

Haixu Wu [*1] Minghao Guo [*1] Zongyi Li [1] Zhiyang Dou [1] Mingsheng Long [2] Kaiming He [1] Wojciech Matusik [1]

当数据已经很贵，如何通过预训练让模型获得可迁移的表示能力

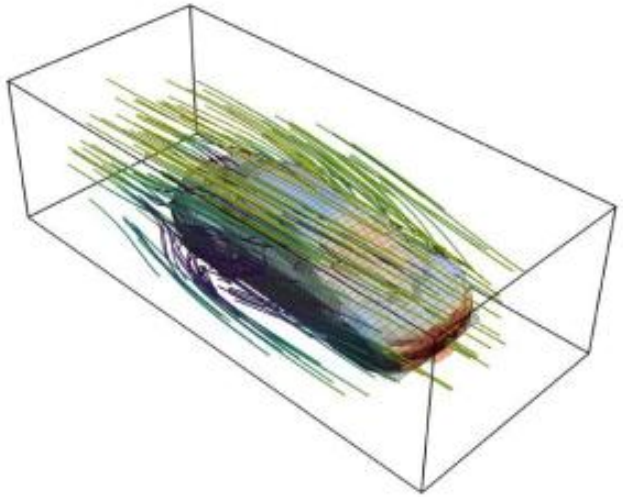
GeoPT：工业级精度场景，数值求解标签数据昂贵



$$\mathcal{L}^{\text{physics}} = \mathbb{E}_{\mathcal{D}} \left[\| \mathcal{F}_\theta(\mathbf{x}; G, S) - \boldsymbol{u}(\mathbf{x}) \|_2^2 \right].$$

模型预测：给定几何 G、物理场条件（流速、方向等）S，在点 x处的物理量

真实物理量（数值求解器算出来的）



在 DrivAerML 空气动力学数据集里，生成一个工业级精度的样本，需要消耗约 6.1×10^4CPU-hours

$$\mathcal{L}^{\text{physics}} = \mathbb{E}_{\mathcal{D}}\left[\left\|\mathcal{F}_\theta(\mathbf{x}; G, S) - \boldsymbol{u}(\mathbf{x})\right\|_2^2\right].$$

Physics **foundation** models.

- Poseidon：预训练流体动力学模型。
- DPOT：扩展到多种物理场，并进行自回归预测。
- Unisolver：将 PDE 信息引入条件建模。
- P3D：扩展到三维流体问题。

通常依赖特定物理领域的数据

只有几何数据?



Self-Supervised Pre-Training

$$\hat{\theta} = \text{Pretrain}(\mathcal{G})$$

$$\theta_0 = \hat{\theta}, \qquad \theta^* = \text{FineTune}\left(\theta_0, \mathcal{D}\right).$$

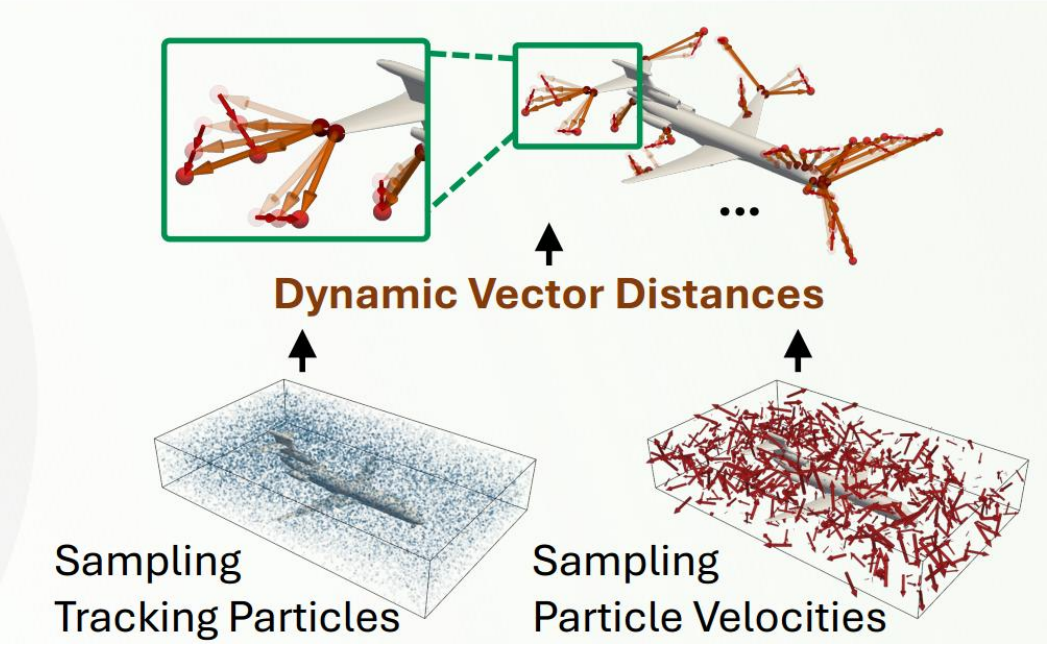
直接对几何自监督学习**不够甚至损害**（只有静态）：

- 视觉：恢复被遮挡的内容
- 几何：给定空间坐标预测几何特征
- occupancy、SDF、VDF 等几何特征

$$\mathcal{L}^{\text{pre}}_{\text{native}} = \mathbb{E}_{x,G}\left[\left\|\hat{F}_\theta(x; G) - h_G(x)\right\|_2^2\right]$$

Intuition：GeoPT 的核心：将系统条件 S转换为合成动力学V

几何形状 + 人工构造的运动条件，学会“几何+动态”通用表示



合成动力学方程直线运动：

$$\frac{dx_t}{dt} = v\,\mathbf{1}_G(x_t), \qquad x_0 = x.$$

时间 t	粒子位置 x_t	状态
0	$(1,5)$	开始运动
1	$(2,5)$	继续运动
2	$(3,5)$	继续运动
3	$(4,5)$	碰到边界
4	$(4,5)$	停止
5	$(4,5)$	停止

$$(x,\ G,\ V) \longrightarrow h_G(x_{0:\tau})$$

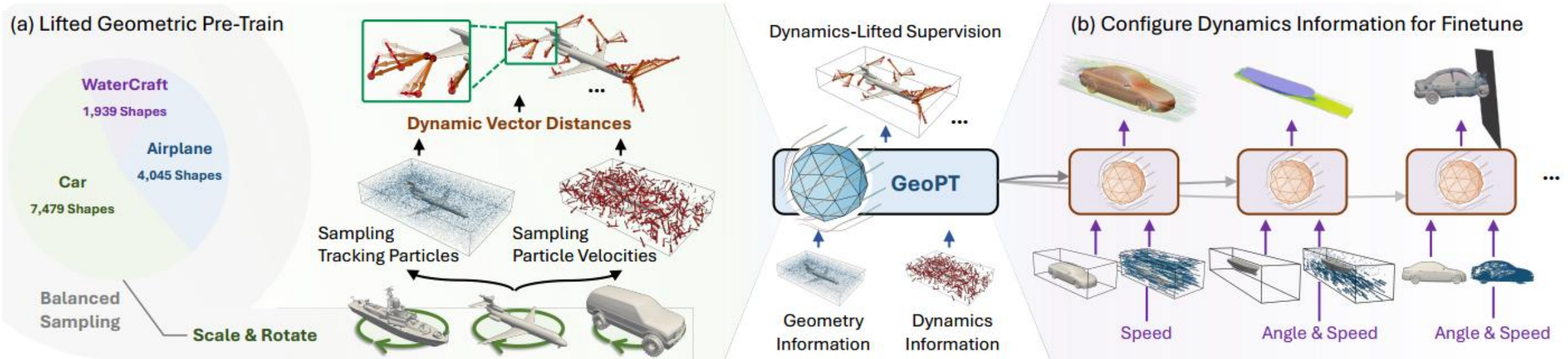
SDF：点离表面有多远（3，2，1，0，0）

VDF：离表面多远，以及表面在哪个方向

$$h_G(x_0) = (3,4,0),$$
$$h_G(x_1) = (2,4,0),$$
$$h_G(x_2) = (1,4,0).$$

目标张量：$Y \in \mathbb{R}^{N\times(\tau+1)\times 3}$,

GeoPT



$$\mathcal{L}_{\text{lifted}}^{\text{pre}} = \mathbb{E}_{x,G,V}\left[\left\|\hat{F}_\theta(x;G,V) - h_G(x_{0:\tau})\right\|_2^2\right]$$

$$\mathcal{L}_{\text{fine}} = \mathbb{E}_{x,G,V_S}\left[\left\|F_\theta(x;G,V_S) - u(x)\right\|_2^2\right]$$

在下游任务中，构造对应的任务特定速度条件 Vs（例如汽车绕流固定V就不用再输入Vs），再使用真实物理场 u(x) 进行微调

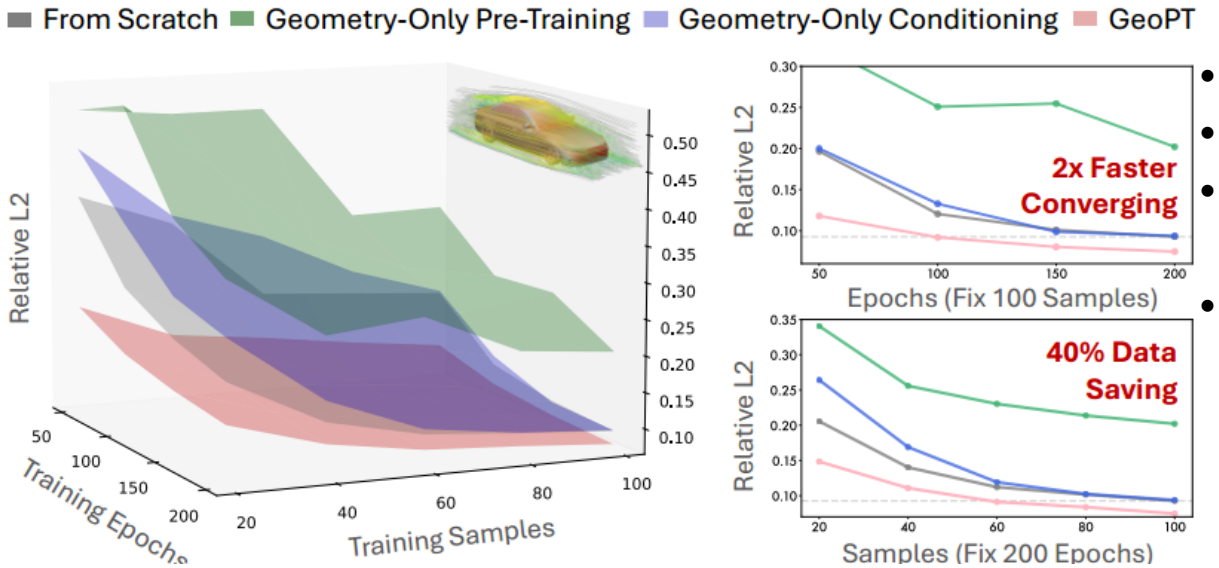
Implement Details

指标	论文报告值
单个几何—动力学样本的点数	36,864
标签计算时间	约 0.2 秒
计算资源	80 个 CPU 核心
相比工业级 CFD 仿真	约快 10^7 倍
离线预计算数据集规模	约 5 TB
预计算耗时	约 3 天

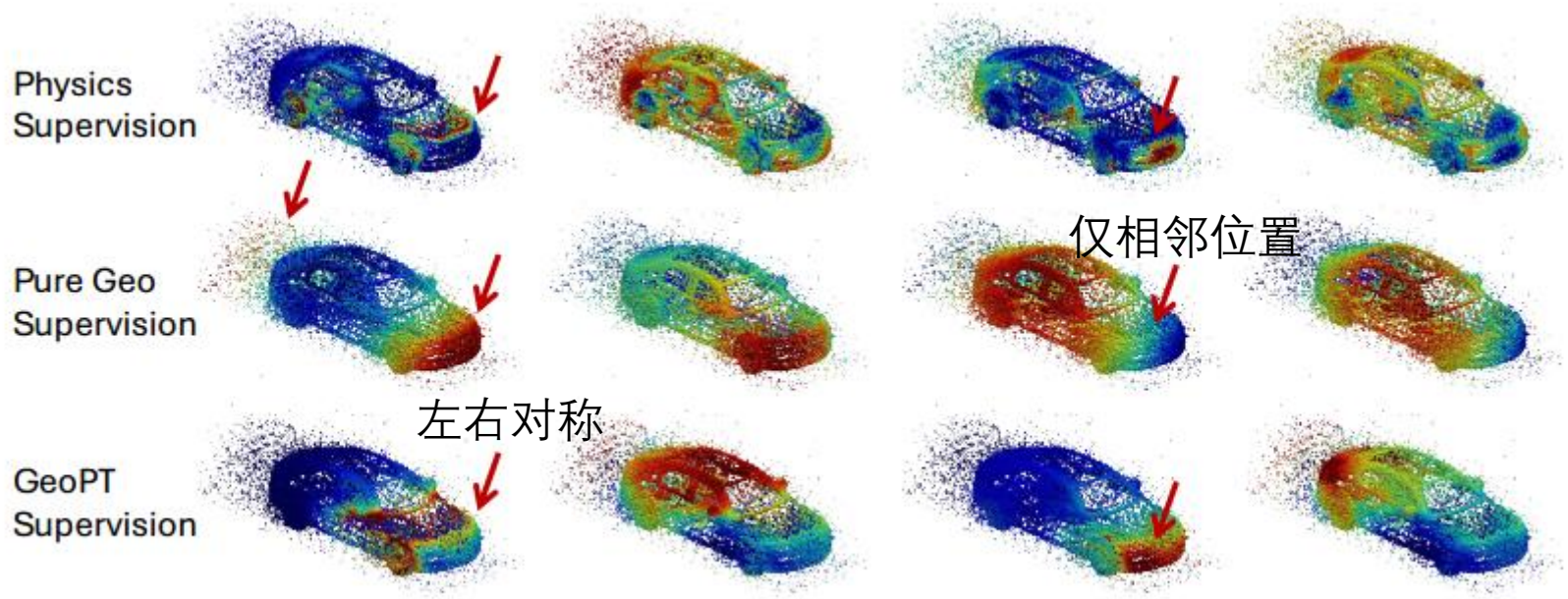
TYPE	BENCHMARKS	#MESH	#VARIABLE	#SIZE
AERO-DYNAMICS	DRIVAERML	∼160M	GEOMETRY	∼6TB
	NASA-CRM	∼450K	GEO, SPEED, AOA	∼3GB
	AIRCRAFT	∼330K	GEO, SPEED, AOA, SLIP	∼7GB
HYDRO-	DTC HULL	∼240K	GEO, YAW ANGLE	∼2GB
CRASH	CAR-CRASH	∼1M	IMPACT ANGLE	∼8GB

To mimic the industrial practice (Ansys Inc., 2026), we adopt or generate ∼100 training samples for each benchmark and test neural simulators on the other 20–50 samples.

Experiment Results

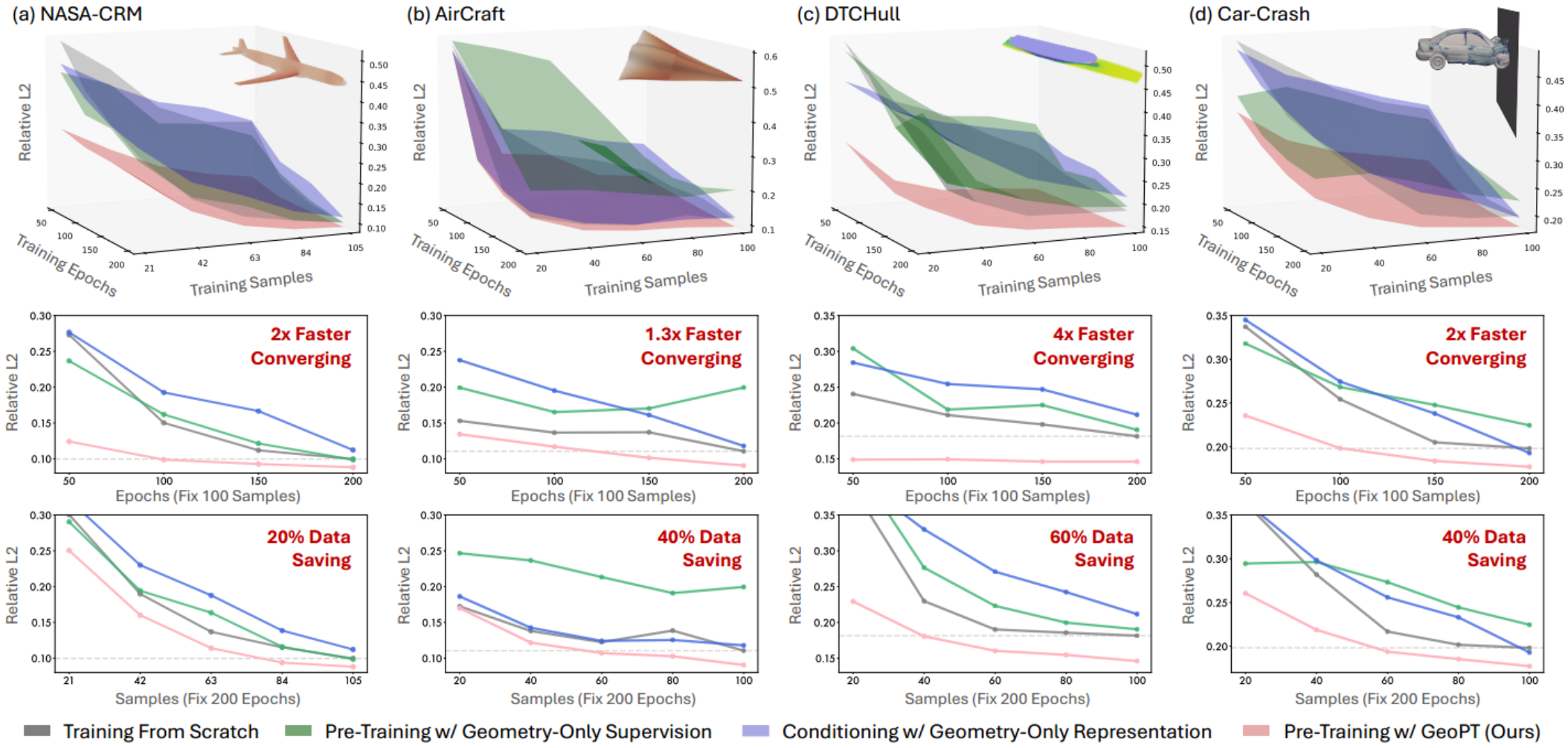


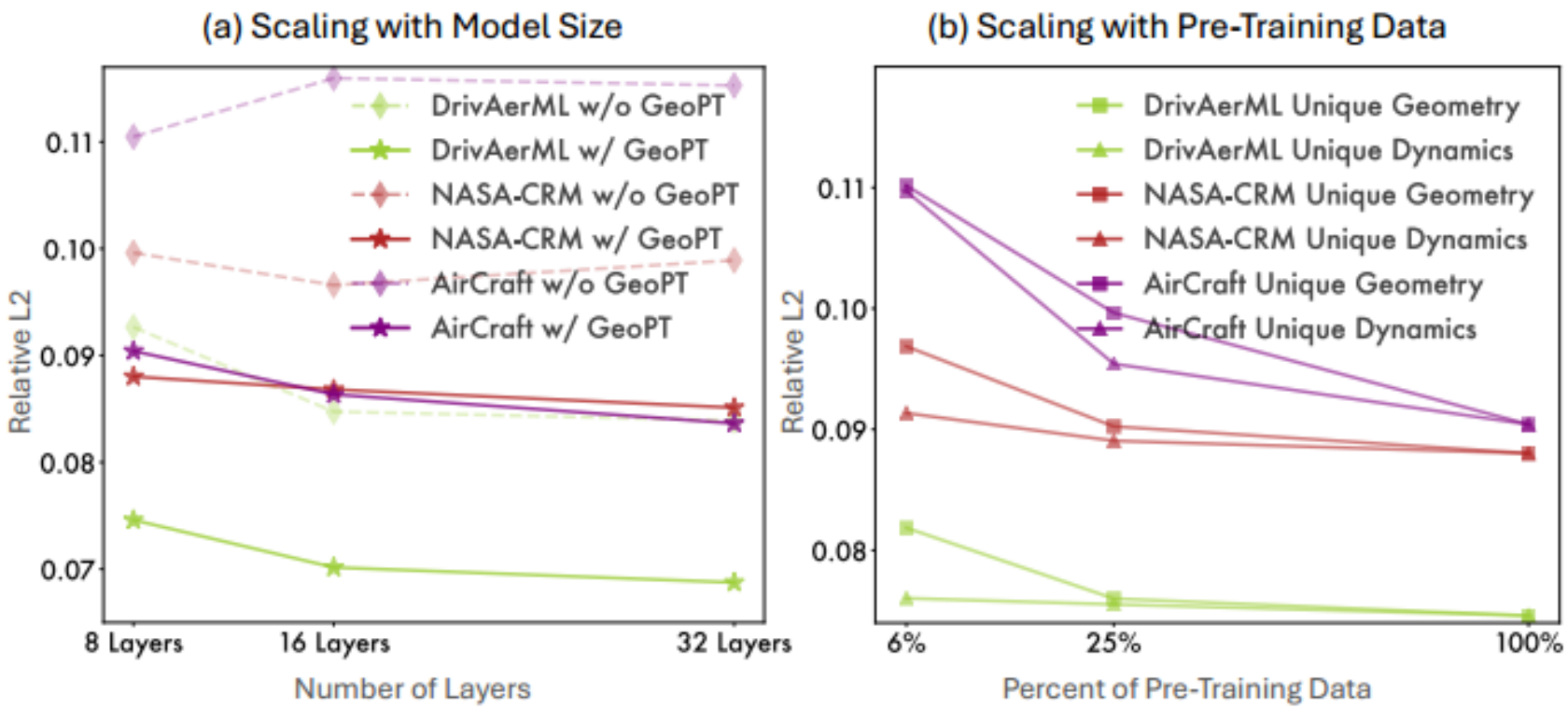
- 灰色：不使用预训练，从头训练。
- 绿色：：只用静态几何特征做预训练。
- 蓝色：使用几何表示作为条件，但不是 GeoPT 的轨迹式预训练。
- 粉色：使用 GeoPT 的合成动力学轨迹和几何特征进行预训练。



(a) Learned Correlation Visualization

Experiment Results





某种正则化

增加预训练数据通常会降低误差。

GeoPT 预训练后，究竟学到了什么，以及为什么这些表示能迁移到真实物理模拟？。

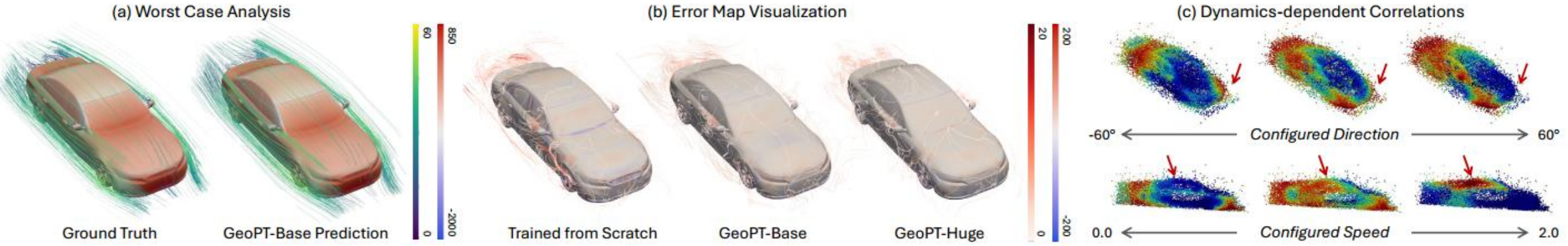


Figure 7. Simulation results and learned representations from GeoPT, including (a) visualization of the prediction results with the worst relative L2 performance in DrivAerML, (b) the error map of surface pressure and surrounding velocity, (c) correlations learned by pre-trained GeoPT under varied dynamics information, such as different directions and speeds of V_S. See Appendix D for more results.

预训练表示不仅对方向条件敏感，也能够响应速度幅值的变化。

Pre-trained GeoPT

保持速度设置在相应实验条件下，改变 V_S 的方向。

- **Configured Direction**：改变 V_S 的方向，从 -60° 到 60°。
- **Configured Speed**：改变 V_S 的速度幅值，从 0.0 到 2.0。

能否成为Physical Foundational Model

在光子传输领域，预训练仍然比直接从原始场数据学到的要好

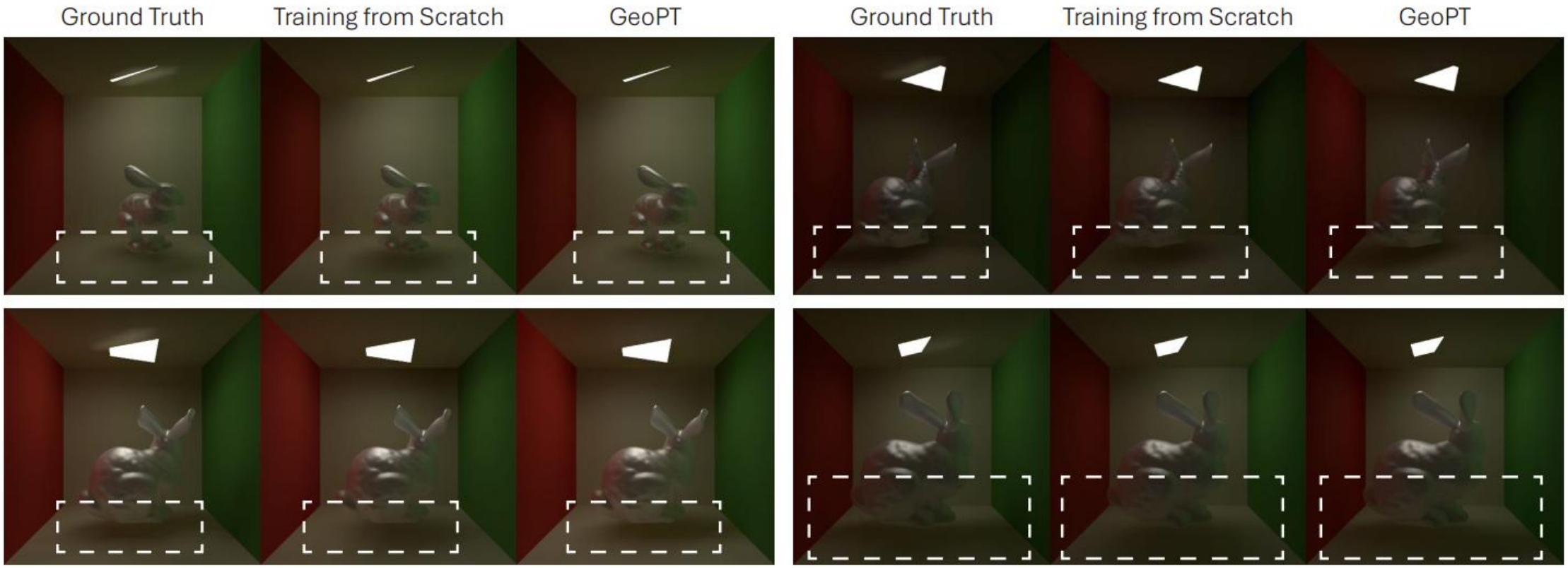
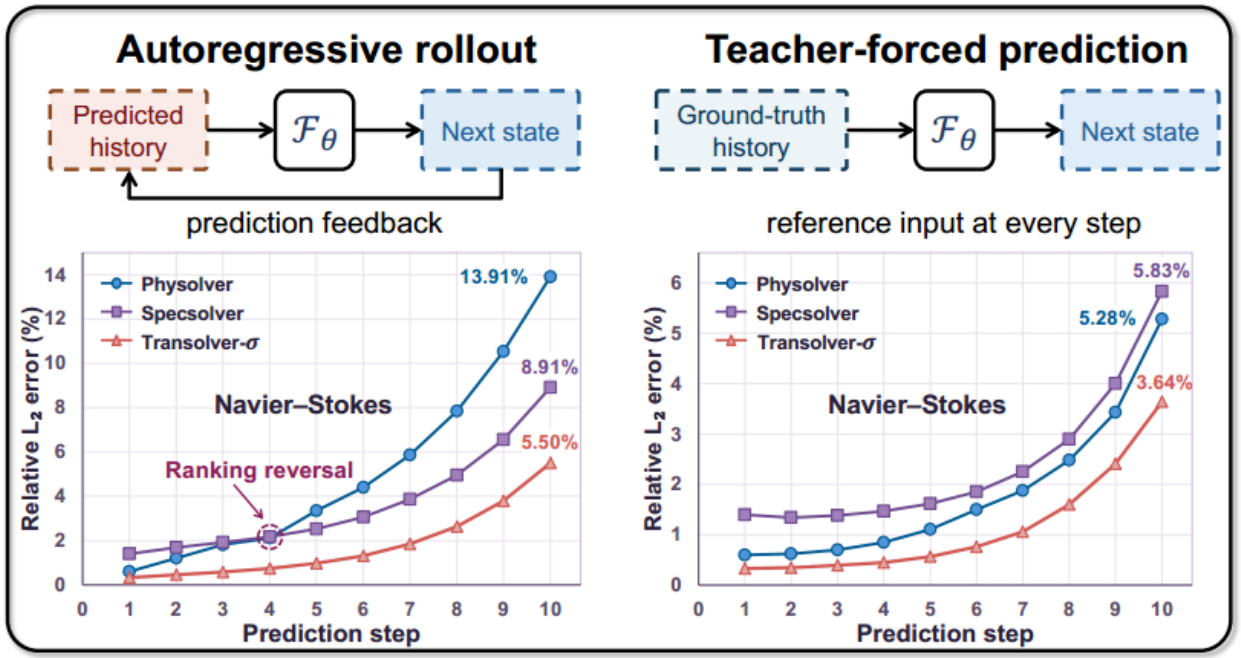


Figure 9. Experiments of neural radiosity simulation on the Cornell box (Goral et al., 1984) with an unseen Stanford bunny geometry.

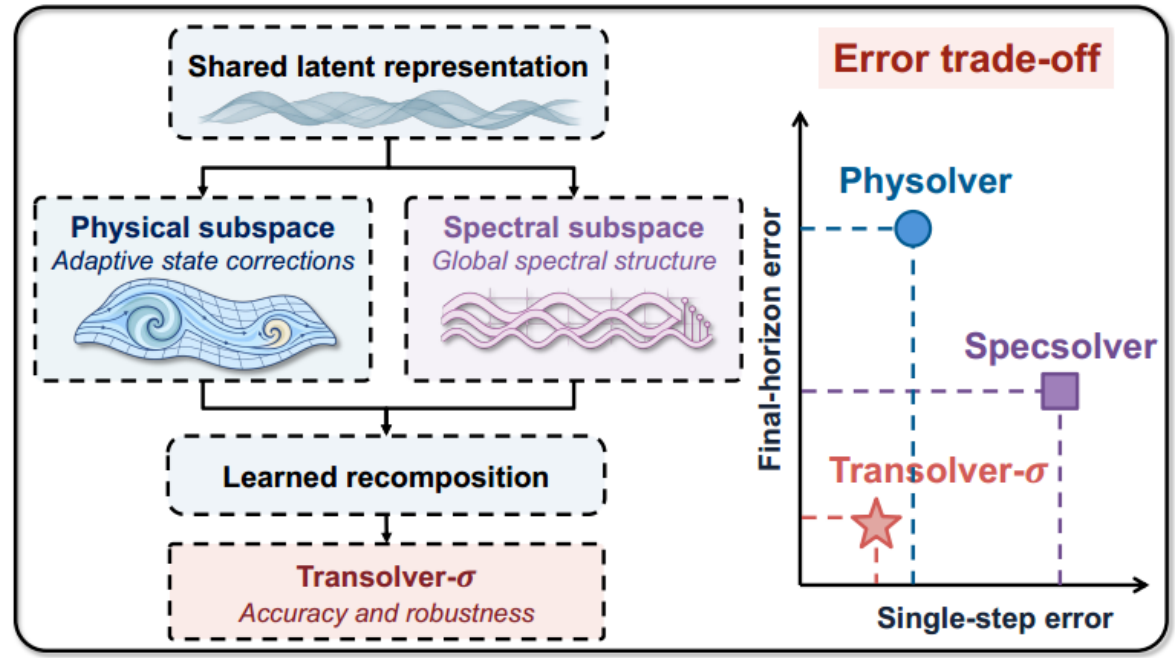
光线传播——粒子速度轨迹

TRANSOLVER-σ: JOINT SPECTRAL-PHYSICAL SUBSPACE MODELING FOR NEURAL PDE SOLVING

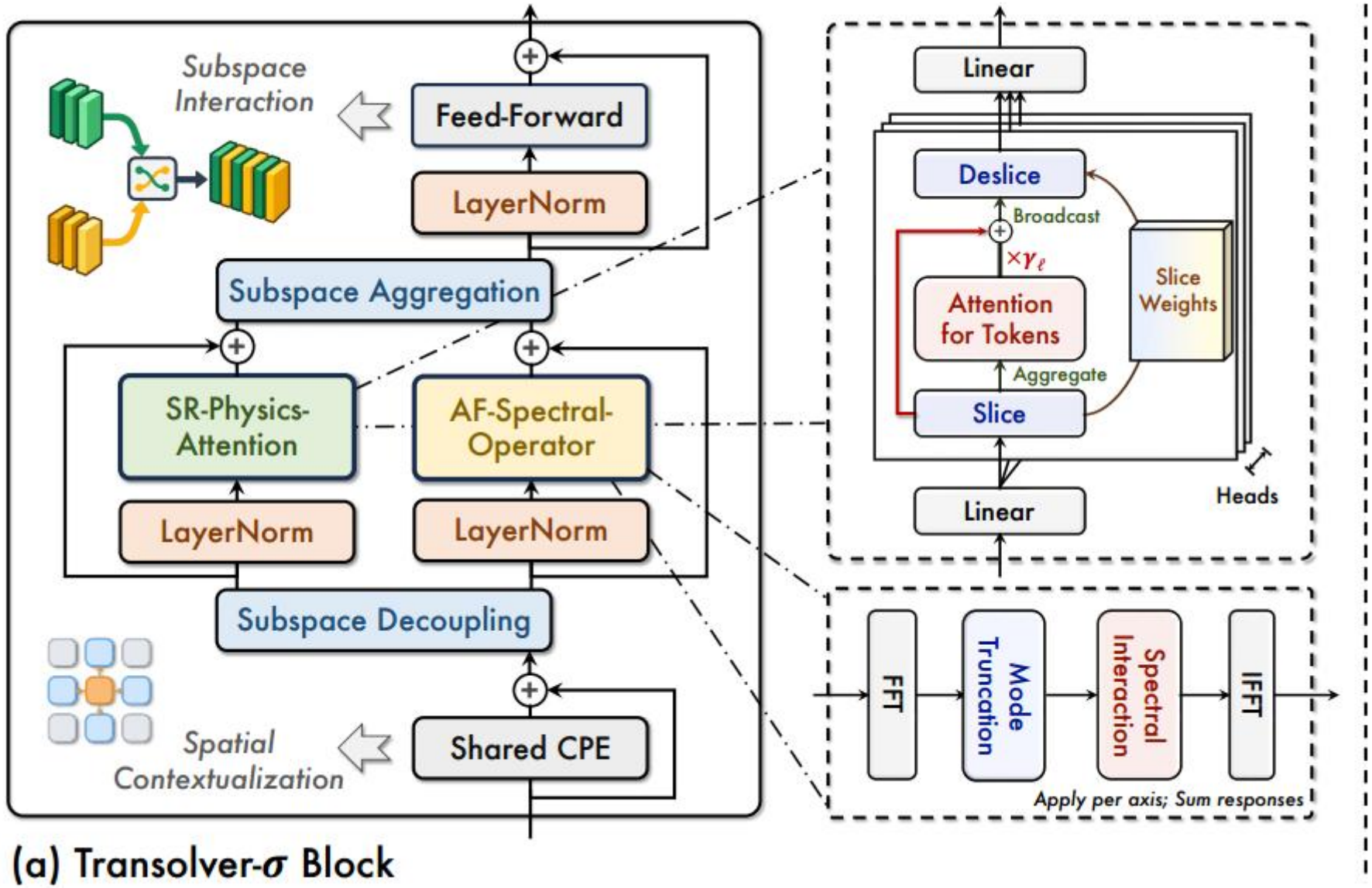
Haonan Shangguan[1,*], **Hang Zhou**[1,*], **Haixu Wu**[2], **Yuezhou Ma**[1], **Jianmin Wang**[1], **Mingsheng Long**[1,†]



(a) Single-step accuracy does not ensure rollout robustness



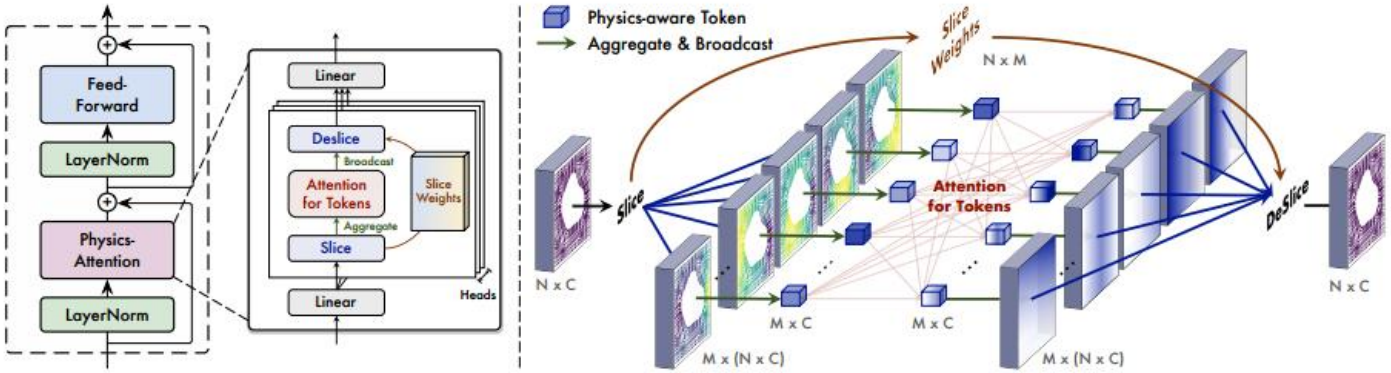
(b) Joint spectral-physical modeling for balanced error trade-off



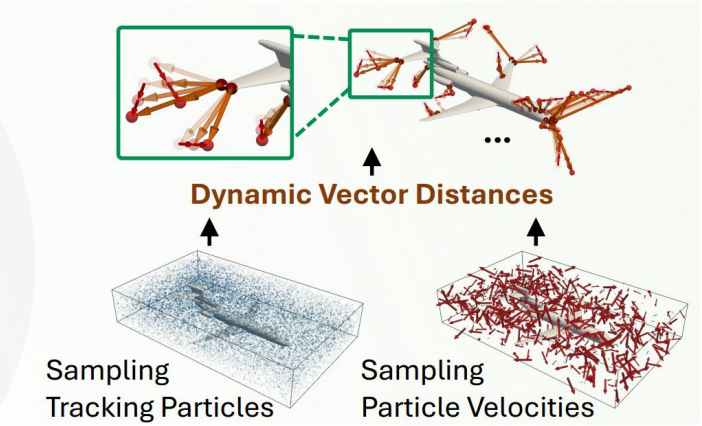
(a) Transolver-σ Block

Table 2: **Results on RealPDEBench.** RMSE, relative L^2, and Fourier-space RMSE (fRMSE) across four real-world measurement tasks. Params denotes the mean parameter count (M) across tasks.

Models	Params (M)	Controlled Cylinder			FSI			Foil			Combustion		
		RMSE	Rel L^2	fRMSE	RMSE	Rel L^2	fRMSE	RMSE	Rel L^2	fRMSE	RMSE	Rel L^2	fRMSE
U-Net	23.08	<u>0.80</u>	<u>5.55</u>	0.10	<u>0.85</u>	<u>5.83</u>	<u>0.07</u>	<u>1.00</u>	<u>1.59</u>	0.11	2.16	54.87	0.26
CNO	8.00	0.81	5.83	<u>0.09</u>	1.05	7.41	0.09	1.36	2.53	0.18	2.48	60.30	0.32
DeepONet	3.53	3.09	23.99	0.58	3.50	25.02	0.51	2.22	3.63	0.28	2.29	57.51	0.28
FNO	109.10	0.97	7.23	0.12	1.29	8.92	0.12	1.30	2.28	0.15	2.26	56.64	0.27
WDNO	233.15	1.15	9.27	0.14	1.17	8.17	0.11	1.62	4.44	0.18	3.80	86.05	0.55
MWT	2.90	1.02	7.47	0.11	1.28	9.10	0.10	1.33	2.27	0.15	2.21	55.60	0.27
GK-Transformer	59.20	1.03	7.67	0.12	1.27	9.16	0.11	1.42	2.45	0.18	2.47	60.83	0.31
Transolver	4.30	1.71	12.00	0.24	2.36	15.67	0.27	2.20	3.70	0.27	3.75	78.25	0.50
DPOT-S-FT	38.73	0.84	5.98	0.10	1.05	7.46	0.08	1.06	1.66	<u>0.10</u>	2.09	53.49	<u>0.24</u>
DPOT-L-FT	667.75	0.84	6.03	0.10	0.99	6.87	0.08	1.05	<u>1.59</u>	0.11	<u>2.08</u>	<u>53.31</u>	<u>0.24</u>
Transolver-σ	3.13	**0.65**	**5.20**	**0.06**	**0.74**	**5.55**	**0.06**	**0.47**	**1.40**	**0.08**	**1.86**	**52.86**	**0.18**



① Mesh → physics ② Attention (Integral) ③ Physics → Mesh



- 自适应物理信息聚合
- 不规则网络（连续物理场表示）
- 计算开销

- **Foundation**

- **Intuition**
- **Framework**